A Structural Solution to the Birch and Swinnerton-Dyer Role-Compression Problem

Analytic Rank, Mordell–Weil Rank, and Bridge Discipline under AASC/UEAP

Amos Jay Maley

June 2026

Proof-class, claim-class, and review locks

Proof Class Lock. This manuscript is a structural role-compression closure proof. It is not a classical arithmetic-geometric proof of BSD, not a proof that analytic rank equals Mordell–Weil rank for all elliptic curves, not a proof of finiteness of $\mathrm{Sha}(E)$, not a rational-point construction, and not a new descent, Euler-system, Iwasawa, or L-function estimate. A classical proof of BSD would be a bridge-completion theorem inside the decompressed architecture identified here.

Claim-Class Lock. The word *solution* in this paper means AASC structural solution of the BSD role-compression problem, i.e. structural closure of the compressed claim architecture induced by the classical BSD assertion. It does not mean that the classical theorem

$$\mathrm{ord}_{s=1} L(E,s) = \mathrm{rank}\, E(\mathbb{Q})$$

has been proved by arithmetic geometry. The classical theorem is preserved as universal analytic-arithmetic bridge completion.

Wrong-mode boundary. A referee demanding a point-construction proof, a proof of $\mathrm{Sha}(E)$ finiteness, or a proof of the leading-coefficient formula is asking for classical bridge completion. That is a legitimate mathematical demand, but it is not the claim class of this manuscript. Same-mode review asks whether the BSD assertion is role-compressed, whether the roles are separated, whether unbridged transitions are exhausted and blocked, and whether the bridge-completion boundary is correctly located.

Nonclaim 0.1 (Mode boundary)**.** This paper does not prove the classical BSD equality, does not prove the refined BSD formula, does not prove new BSD cases, and does not claim that AASC generates missing arithmetic data. Its theorem is

$$\mathrm{RoleSolution}_{\mathrm{AASC}}(\text{BSD-RoleCompression}),$$

namely structural closure of the BSD role-compression problem.

Mode declaration. This paper is written in AASC mode. AASC is a constraint formalism, not an analytic-number-theoretic, arithmetic-geometric, descent-theoretic, or point-construction method. It does not attempt to prove the Birch and Swinnerton-Dyer Conjecture by constructing rational points, proving finiteness of the Tate–Shafarevich group, deriving a new Euler-system theorem, proving a new

Iwasawa main conjecture, or establishing the leading-coefficient formula by conventional arithmetic geometry. It proves a structural solution to the Birch and Swinnerton-Dyer role-compression problem by showing that the classical formulation compresses distinct standing-bearing roles: elliptic-curve carrier, analytic L-function carrier, vanishing-order standing, Mordell–Weil rank readout, and the analytic-arithmetic bridge governing the transition among them.

A classical arithmetic-geometric proof of the Birch and Swinnerton-Dyer Conjecture itself would be a bridge-completion theorem inside the decompressed architecture identified here. It would not be a rival to this paper's structural solution. AASC does not compete with an analytic or arithmetic proof of rank equality; it determines the role architecture in which such a proof has standing.

The classical Birch and Swinnerton-Dyer assertion is classically well formed as a mathematical proposition. In AASC mode, however, the same assertion is structurally under-typed when left compressed. It is not AASC structurally well posed as a single-role question until the elliptic-curve carrier, analytic vanishing-order standing, arithmetic rank readout, and analytic-arithmetic bridge are separated.

Plain-language compression. In ordinary terms, the Birch and Swinnerton-Dyer Conjecture says that the order of vanishing of an elliptic curve's L-function at $s = 1$ equals the rank of the group of rational points. This paper does not prove that equality by constructing points or controlling descent groups. It shows that the sentence hides a transition from one kind of status, analytic vanishing order, to another kind of status, Mordell–Weil group rank. The three-layer bridge used below separates the place where that transition must occur, the admissibility conditions any proposed transition must satisfy, and the stronger classical proof burden required to complete the transition. AASC makes that transition explicit and proves what cannot legitimately substitute for the analytic-arithmetic bridge.

Abstract

This paper gives a structural AASC solution to the Birch and Swinnerton-Dyer role-compression problem induced by the classical BSD assertion. For an elliptic curve $E/\mathbb{Q}$, the classical rank assertion is

$$\operatorname{ord}_{s=1} L(E,s) = \operatorname{rank} E(\mathbb{Q}).$$

The refined conjecture relates the leading coefficient of $L(E,s)$ at $s = 1$ to the regulator, real period, Tamagawa factors, torsion subgroup, and Tate–Shafarevich group. These assertions are classically meaningful as arithmetic-geometric targets, but they are structurally compressed when read as a single undifferentiated report: analytic L-function vanishing is presented as if it directly occupied the arithmetic Mordell–Weil rank-readout role.

The central neutral statuses used here are

$$\operatorname{Van}(E,n) \Longleftrightarrow \operatorname{ord}_{s=1} L(E,s) = n$$

and

$$\operatorname{RankRead}(E,n) \Longleftrightarrow \operatorname{rank} E(\mathbb{Q}) = n.$$

The analytic-arithmetic bridge is separated into three layers: the bridge slot $B^{\mathrm{slot}}_{\mathrm{BSD}}(E,n)$, bridge admissibility $B^{\mathrm{adm}}_{\mathrm{BSD}}(E,n)$, and bridge completion $B^{\mathrm{comp}}_{\mathrm{BSD}}(E,n)$. The first two layers type and govern the transition without containing the rank conclusion; the third is the classical arithmetic bridge-completion burden. The refined leading-coefficient formula is treated as full bridge-completion architecture, not as a silent replacement for rank readout.

The paper proves, in proof order, that the classical BSD assertion role-compresses elliptic-curve carrier, analytic L-function carrier, vanishing-order standing, Mordell–Weil rank readout, and analytic-arithmetic bridge; that these roles are distinct; that unbridged promotion from analytic rank to arithmetic rank is inadmissible; that surrogate structures such as Selmer groups, Tate–Shafarevich data, Heegner points, regulators, numerical L-function approximations, Iwasawa-theoretic data, or special-case theorems cannot silently occupy the Mordell–Weil rank-readout role; that no same-scope totalization of rank equality is available through selector import, repair, carrier transfer, descent substitution, numerical extrapolation, or local-to-global factorization; and that rank readout is role occupation rather than preferred-generator selection.

The proof uses the AASC kernel of reference, standing, admissibility, and irreversibility; UEAP report discipline; the AMetric no-selector boundary; same-scope operator closure; no-generators/no-carriers/no-repairs; no post-selection repair; prohibition of silent redescription; non-factorizability of admissibility; boundary-trace fixation; and role-occupancy closure. These imported results are restated in local BSD form before use. No imported result is used to prove analytic continuation of $L(E, s)$, rational-point existence, finiteness of $\mathrm{Sha}(E)$, or the full leading-coefficient formula. They are used to classify standing, reportability, carrier/readout separation, bridge occupation, and illicit status transitions.

The conclusion is that the BSD role-compression problem is structurally solved in AASC mode, i.e. $\mathrm{RoleSolution}_{\mathrm{AASC}}(\text{BSD-RoleCompression})$ holds. The analytic-arithmetic endpoint is preserved as universal bridge completion: a proof in classical arithmetic geometry would show, for every elliptic curve $E/\mathbb{Q}$, that the analytic vanishing-order report and Mordell–Weil rank readout coincide, and that the refined leading coefficient is governed by the standard BSD bridge factors. Such a proof would instantiate the bridge identified here rather than refute the structural solution.

Keywords. Birch and Swinnerton-Dyer Conjecture; elliptic curves; L-functions; analytic rank; Mordell–Weil rank; Tate–Shafarevich group; Selmer groups; regulator; AASC; UEAP; admissibility; standing; role compression; structural solution; analytic-arithmetic bridge; no silent redescription; no post-selection repair.

How to read this paper

Classical BSD readers should begin with the Proof Class Lock, Claim-Class Lock, and Endpoint Discipline table before evaluating the theorem statements. The manuscript does not claim to complete the classical arithmetic bridge. Its central theorem is the structural closure statement

$$\mathrm{RoleSolution}_{\mathrm{AASC}}(\text{BSD-RoleCompression}),$$

while the classical rank equality and refined leading-coefficient formula are preserved as bridge-completion targets. The bridge-completion corollary states the exact relation: a future arithmetic proof of BSD would instantiate the bridge located here, not compete with the structural role-compression result.

Endpoint discipline

This manuscript distinguishes five endpoints that are often collapsed in discussion. The first is the familiar classical BSD rank endpoint. The second is the refined leading-coefficient endpoint. The third is the structural role-compression endpoint. The fourth is the no-unbridged-promotion endpoint. The fifth is the bridge-completion boundary for future arithmetic-geometric proofs.

Endpoint	Statement	Status
Classical rank endpoint	For every elliptic curve $E/\mathbb{Q}$, $\mathrm{ord}_{s=1} L(E, s) = \mathrm{rank}\, E(\mathbb{Q})$.	Preserved as analytic-arithmetic bridge completion.

Refined leading-coefficient endpoint	The first nonzero Taylor coefficient of $L(E, s)$ at $s = 1$ is given by the BSD product of regulator, real period, Tamagawa factors, torsion denominator, and $\lvert \mathrm{Sha}(E) \rvert$ when the terms are standing-fixed.	Preserved as full bridge completion.
Structural role-compression endpoint	The classical BSD assertion compresses elliptic-curve carrier, analytic standing, arithmetic readout, and bridge.	Solved here in AASC structural mode.
No-unbridged-promotion endpoint	The report $\mathrm{ord}_{s=1} L(E, s) = n$ may not be promoted to $\mathrm{rank}\, E(\mathbb{Q}) = n$ without bridge occupation.	Proved in this paper.
Role-occupation endpoint	Mordell–Weil rank readout is lawful occupation of the free-rank role, not selection of preferred generators or bases.	Proved in this paper.
Descriptive mismatch endpoint	A statement such as $\mathrm{ord}_{s=1} L(E, s) \neq \mathrm{rank}\, E(\mathbb{Q})$, if established, is legitimate mathematical data and not illicit by itself.	Preserved.
Analytic proof boundary	A future arithmetic-geometric proof would prove universal bridge completion for every $E/\mathbb{Q}$.	Located inside the solved structure.

Rank, refined, and structural layers.

BSD layer	What it bridges	Completion burden
Rank BSD	Analytic order of vanishing $\leftrightarrow$ Mordell–Weil free rank.	Prove equality of analytic and arithmetic ranks for the target curve.
Refined BSD	Leading coefficient $\leftrightarrow$ period, regulator, Tamagawa, torsion, and Sha product.	Prove rank equality, Sha finiteness/order, regulator conventions, period, Tamagawa, and torsion normalizations.
AASC structural BSD	Carrier/standing/readout/bridge typing.	Prove role compression, no unbridged promotion, surrogate exclusion, totalization block, and bridge-location discipline.

Nonmembership and mismatch clarification. The paper does not say that a descriptive mismatch statement

$$\mathrm{ord}_{s=1} L(E, s) \neq \mathrm{rank}\, E(\mathbb{Q})$$

is illicit. If established, it would be a legitimate mathematical statement. What AASC blocks is the same-scope promotion of analytic standing, surrogate structure, numerical evidence, Selmer standing, obstruction data, or neighboring carrier data into Mordell–Weil rank readout without bridge occupation of the arithmetic-readout role.

Nonclaim 0.2 (Mode boundary)**.** This paper is not an analytic or arithmetic proof of BSD in the classical sense. It does not claim to prove the equality $\mathrm{ord}_{s=1} L(E, s) = \mathrm{rank}\, E(\mathbb{Q})$ for every elliptic curve by conventional arithmetic geometry. It does not construct a rank-generating set of rational points, prove finiteness of $\mathrm{Sha}(E)$, or prove the leading-coefficient formula. Its claim is that BSD has a structurally compressed analytic/arithmetic carrier-standing-readout architecture and that AASC closes that structural problem by role separation, transition discipline, and exhaustion of unbridged routes.

Contents

1 The classical BSD assertion as role compression

1.1 The classical assertion retained

Let $E/\mathbb{Q}$ be an elliptic curve. The Hasse-Weil L-function of E is denoted by $L(E, s)$. The Mordell–Weil theorem says that the abelian group $E(\mathbb{Q})$ is finitely generated, so

$$E(\mathbb{Q}) \cong E(\mathbb{Q})_{\mathrm{tors}} \oplus \mathbb{Z}^r$$

for a unique integer $r = \operatorname{rank} E(\mathbb{Q})$. The rank part of the Birch and Swinnerton-Dyer Conjecture asserts

$$\operatorname{ord}_{s=1} L(E, s) = \operatorname{rank} E(\mathbb{Q}). \tag{1}$$

This assertion is retained. It is not dissolved into vocabulary. It is the classical arithmetic-geometric endpoint.

The refined conjecture further relates the leading coefficient of $L(E, s)$ at $s = 1$ to a product of arithmetic invariants. In one standard informal shape, if $r = \operatorname{rank} E(\mathbb{Q})$, then

$$\frac{L^{(r)}(E, 1)}{r!} = \frac{\Omega_E \ \mathrm{Reg}_E \ |\mathrm{Sha}(E)| \prod_p c_p}{|E(\mathbb{Q})_{\mathrm{tors}}|^2}, \tag{2}$$

with the usual interpretive qualifications concerning periods, Tamagawa factors, regulator, torsion, and Tate–Shafarevich group. This paper also preserves that refined endpoint. It treats it as full bridge-completion architecture rather than as a shortcut from analytic standing to rank readout.

Remark 1.1 (Classically well formed but not structurally single-role)**.** Equation (1) is classically well formed as a mathematical sentence. The structural claim is different. As a single compressed equality, it is not AASC structurally well posed until the analytic vanishing-order role, the Mordell–Weil rank role, and the bridge condition between them are separated.

1.2 Why the equality form is structurally dangerous

The equality sign in (1) is harmless in classical mathematical notation when read as the conjectural endpoint. It is dangerous only if it is read as a same-role continuation. The left side is an analytic invariant extracted from the behavior of an L-function at a critical point. The right side is an arithmetic invariant of a finitely generated abelian group of rational points. The conjecture says these two statuses coincide. It does not make them the same role.

AASC asks whether a problem statement presents one typed role or compresses several roles into one assertion. For BSD the answer is that the classical equality is role-compressed. It moves from analytic L-function standing to Mordell–Weil rank readout without displaying the bridge whose occupation is required for that transition.

1.3 Formal definition of role compression

Definition 1.2 (Standing-bearing role)**.** A standing-bearing role is a typed position in a claim architecture whose occupant determines what kind of status the claim can bear. A role may be a carrier role, a standing role, a readout role, a bridge role, an obstruction role, or a reportability role. Roles are distinct when changing one while preserving the others changes the strongest reportable status under the fixed audit basis.

Definition 1.3 (Role-compressed problem statement)**.** Let P be a problem statement and let $R_1, \ldots, R_n$ be standing-bearing roles. We write

$$\mathrm{RoleCompresses}(P; R_1, \ldots, R_n)$$

when the following neutral typing conditions hold:

1. P contains a transition among the roles R_i, rather than merely a statement internal to a single role.
2. P presents that transition as a single undifferentiated assertion, implication, equality, or equivalence.
3. At least one carrier, standing, readout, bridge, or reportability condition required for status-preserving transition is not explicitly displayed in P.
4. The omitted condition changes reportability under UEAP: omitting it allows a stronger report than the currently fixed carrier alone supports.

This definition is neutral. It does not itself declare the compression illicit, and it does not pre-build a failure verdict into the concept of compression. It only identifies an under-typed transition among roles.

Definition 1.4 (Laundering-risk compression)**.** A role compression has laundering risk when the omitted bridge or reportability condition opens at least one same-scope overreporting route: unbridged promotion, surrogate substitution, carrier transfer, post-selection repair, silent redescription, selector import, or factorized local admissibility.

Definition 1.5 (Structural role-compression solution)**.** A structural AASC solution of a role-compressed problem P is a proof of the following data:

1. the compressed roles are typed and separated;

2. the admissible transition conditions among them are stated;
3. unlicensed transitions are exhausted and blocked;
4. the exact bridge condition required for the stronger report is identified;
5. the analytic, arithmetic, or geometric endpoint, if any, is re-expressed as bridge completion inside the separated role architecture.

We write $\mathrm{RoleSolution}(P)$ for this package.

1.4 Why the structural solution criterion is not stipulative

The criterion above is the fixed form of an AASC structural solution. AASC does not solve by analytic generation; it solves by closure of typed claim architectures. A role-compression solution must therefore identify the roles, state the admissible transitions among them, block transitions that exceed standing, identify the bridge required for the stronger report, and locate any classical endpoint as bridge completion inside the separated architecture.

These requirements follow from the kernel of reference, admissibility, standing, and irreversibility; from UEAP report preservation; from the AMetric no-selector boundary; from no-repair and no-silent-redescription discipline; and from role-occupancy closure. They are framework requirements, not BSD-specific conveniences. A manuscript that satisfies them has not merely chosen a favorable notion of solution; it has met the AASC criterion for closing a compressed structural problem.

1.5 The BSD roles

Definition 1.6 (Elliptic-curve carrier role)**.** For an elliptic curve $E/\mathbb{Q}$, the elliptic-curve carrier role is

$$R_E(E) = C_E = (E/\mathbb{Q}, E(\mathbb{Q}), L(E,s), \text{local factors}, \text{lawful equivalences}).$$

It fixes the object under audit and the ordinary analytic and arithmetic structures attached to that object.

Definition 1.7 (L-function carrier role)**.** The L-function carrier role is the analytic package

$$C_L(E) = \big(L(E,s), \Lambda(E,s), \text{Euler factors}, \text{functional equation}, s=1, \sim_{\mathrm{an}}\big).$$

For elliptic curves over $\mathbb{Q}$, analytic continuation and the functional equation are fixed in the classical setting through modularity. This manuscript uses those facts only to type the analytic carrier and the vanishing-order report; it does not use modularity to prove Mordell–Weil rank readout.

Definition 1.8 (Analytic vanishing-order standing)**.** For $n \in \mathbb{Z}_{\geq 0}$, define

$$\mathrm{Van}(E,n) \iff \operatorname{ord}_{s=1} L(E,s) = n.$$

This is analytic standing. It reports a status of the L-function carrier. In ordinary BSD terminology, the integer $\operatorname{ord}_{s=1} L(E,s)$ is the analytic rank. In the present AASC proof, the phrase *analytic vanishing-order standing* is used when the emphasis is on report status rather than arithmetic readout.

Definition 1.9 (Arithmetic rank readout)**.** For $n \in \mathbb{Z}_{\geq 0}$, define

$$\mathrm{RankRead}(E,n) \iff \operatorname{rank} E(\mathbb{Q}) = n.$$

Equivalently, $E(\mathbb{Q})/E(\mathbb{Q})_{\mathrm{tors}} \cong \mathbb{Z}^n$. This is arithmetic readout. It reports the free-rank status of the Mordell–Weil group.

Definition 1.10 (BSD bridge-slot layer)**.** For a fixed pair (E, n), the BSD bridge slot is the declared transition position

$$B^{\mathrm{slot}}_{\mathrm{BSD}}(E, n)$$

between analytic vanishing-order standing and Mordell–Weil rank readout. It fixes the tuple

$$(E/\mathbb{Q}, C_L(E), \mathrm{Van}(E, n), E(\mathbb{Q}), r_{\mathrm{rank}}(E, n)),$$

where $r_{\mathrm{rank}}(E, n)$ is the arithmetic readout role whose occupant is the value n of the free rank of $E(\mathbb{Q})$. The bridge slot does not assert $\mathrm{RankRead}(E, n)$; it declares the typed place where a same-target bridge would have to operate.

Definition 1.11 (BSD bridge-admissibility layer)**.** The BSD bridge-admissibility condition

$$B^{\mathrm{adm}}_{\mathrm{BSD}}(E, n)$$

holds for a proposed route only when the route preserves the same elliptic curve $E/\mathbb{Q}$, the same analytic-rank report $\mathrm{Van}(E, n)$, the same Mordell–Weil group carrier, and the same rank-readout slot, while excluding surrogate substitution, post-selection repair, local-to-global overassembly, unlicensed selector authority, and second-classifier governance. Bridge admissibility does not assert the arithmetic conclusion. It says only that a proposed route is typed as a same-scope candidate for bridge completion.

Definition 1.12 (BSD bridge-completion layer)**.** The BSD rank bridge is completed,

$$B^{\mathrm{comp}}_{\mathrm{BSD}}(E, n),$$

when the bridge slot and admissibility conditions are discharged by proof-grade analytic and arithmetic traces for the same target, including a proof-grade Mordell–Weil free-rank certificate

$$E(\mathbb{Q})/E(\mathbb{Q})_{\mathrm{tors}} \cong \mathbb{Z}^n.$$

A classical arithmetic proof of the rank part of BSD would prove $B^{\mathrm{comp}}_{\mathrm{BSD}}(E, n)$ for every target pair (E, n) with $\mathrm{Van}(E, n)$. Bridge completion is therefore the classical endpoint layer. It is not the structural theorem itself; the structural theorem identifies the slot, fixes admissibility discipline, and excludes unbridged substitutes.

Definition 1.13 (BSD bridge certificate package)**.** A completed BSD rank-bridge certificate for a fixed pair (E, n) is a same-scope certificate package

$$\mathcal{B}_{\mathrm{BSD}}(E, n) = (E, n, C_L(E), C_{\mathrm{MW}}(E), T_{\mathrm{an}}, T_{\mathrm{arith}}, \Gamma),$$

where T_{an} is the analytic standing trace, T_{arith} is the arithmetic free-rank trace, and Γ records bridge-admissibility discipline: no retargeting, no surrogate substitution, no post-selection repair, no local-to-global overassembly, no unlicensed selector authority, and no second-classifier governance. The certificate package is not available merely because the bridge slot is named; it is what a bridge-complete proof must supply.

Proposition 1.14 (Three-layer bridge non-circularity)**.** *The slot and admissibility layers $B^{\mathrm{slot}}_{\mathrm{BSD}}(E, n)$ and $B^{\mathrm{adm}}_{\mathrm{BSD}}(E, n)$ do not contain* $\mathrm{RankRead}(E, n)$ *or the classical BSD rank equality by definition. The arithmetic conclusion appears only at bridge-completion level $B^{\mathrm{comp}}_{\mathrm{BSD}}(E, n)$.*

Proof. If the bridge slot contained Mordell–Weil rank equality, BSD would be definitional rather than conjectural. If bridge admissibility contained Mordell–Weil rank equality, no incomplete bridge program could be audited before completion. The slot therefore fixes the typed place of transition, and admissibility fixes the same-scope conditions under which a proposed route may count as a candidate. Only completion supplies the proof-grade arithmetic trace. Thus the conclusion belongs to $B^{\mathrm{comp}}_{\mathrm{BSD}}$, not to $B^{\mathrm{slot}}_{\mathrm{BSD}}$ or $B^{\mathrm{adm}}_{\mathrm{BSD}}$. □

Remark 1.15 (Why the three-layer bridge is not a retreat)**.** This is the same architectural separation used in the RH and Hodge structural papers: the bridge role is distinct from the classical endpoint, and bridge completion is the future classical proof boundary. The present theorem proves slot typing, admissibility discipline, and exclusion of unbridged substitutes. It does not claim to supply universal bridge completion.

1.6 The BSD Role-Compression Lemma

Lemma 1.16 (BSD Role-Compression Lemma)**.** *Let $P = \mathrm{BSD}(E)$ be the classical BSD rank assertion*

$$\operatorname{ord}_{s=1} L(E, s) = \operatorname{rank} E(\mathbb{Q}).$$

Then

$$\mathrm{RoleCompresses}(P; R_E, R_L, R_{\mathrm{Van}}, R_{\mathrm{RankRead}}, R_B).$$

Proof. We verify the clauses in the definition.

First, the assertion contains a transition. The left side is analytic vanishing-order standing: $\operatorname{ord}_{s=1} L(E, s) = n$. The right side is arithmetic rank readout: $\operatorname{rank} E(\mathbb{Q}) = n$. The assertion therefore moves from the analytic carrier to the arithmetic Mordell–Weil carrier.

Second, the roles are distinct. The analytic role is typed by the behavior of $L(E, s)$ at $s = 1$. The arithmetic role is typed by the free abelian rank of the finitely generated group $E(\mathbb{Q})$. These are different carriers and different report statuses.

Third, the compact equality does not separately display the bridge condition. It does not state which analytic-arithmetic data lawfully carry the vanishing-order report into the Mordell–Weil rank-readout role. It states the endpoint equality directly.

Fourth, the omitted bridge condition changes reportability. A report of analytic rank is not already a report of Mordell–Weil rank. Omitting the bridge therefore permits a stronger arithmetic report than the analytic carrier alone supports. Hence the BSD assertion satisfies the neutral definition of role compression. □

Theorem 1.17 (BSD laundering-risk theorem)**.** *The BSD role compression identified above has laundering risk. The omitted bridge condition opens the standard overreporting routes: unbridged promotion of analytic rank to arithmetic rank, surrogate substitution by Selmer or Tate–Shafarevich data, carrier transfer from known cases, numerical-rank overreport, post-selection repair, silent re-description, selector import by preferred generators or height ranks, and factorized local admissibility.*

Proof. If the analytic-arithmetic bridge is not displayed, the transition from $\mathrm{Van}(E, n)$ to $\mathrm{RankRead}(E, n)$ can be attempted by routes other than bridge completion. One may promote analytic rank to Mordell–Weil rank; substitute Selmer rank, Heegner-point data, regulator data, Tate–Shafarevich standing, Iwasawa-theoretic standing, or numerical L-function evidence for free-rank readout; transfer rank equality from rank 0 or rank 1 theorems to a general target curve; treat later point construction as repair of an earlier unbridged rank report; redescribe a surrogate carrier as rank readout; choose a preferred generator set or height-minimal search output by selector; or assemble local factors into a global rank report without a global compatibility certificate. Later sections prove that each route is bookkeeping, bridge completion, explicit scope reset, or inadmissible. Therefore the neutral compression has the stated laundering risk. □

1.7 The structural target

The target of this paper is not the classical proof of the BSD equality for all elliptic curves. The target is

$$\mathrm{RoleSolution}_{\mathrm{AASC}}(\text{BSD-RoleCompression}),$$

where BSD-RoleCompression denotes the compressed claim architecture induced by the classical BSD assertion. This notation is deliberately narrower than RoleSolution(BSD) read in the classical arithmetic-geometric sense. The structural theorem closes the role-compression problem; it does not discharge the universal arithmetic bridge.

The remainder of the paper proves that target in proof order. First, it states the corpus matrix scan and imported AASC constraints in local BSD form. Second, it formalizes the elliptic-curve carrier, analytic standing predicate, arithmetic rank-readout role, bridge-slot schema, and refined bridge architecture. Third, it proves role separation. Fourth, it proves no unbridged promotion, no surrogate readout, no unbridged totalization, and role-occupation closure. The structural BSD solution theorem is stated only after those results have been established.

2 Corpus matrix scan and AASC results used locally

2.1 Matrix scan and source freeze

The support scan for this manuscript used the uploaded `AASC_Corpus_Control_Matrix.csv`. The matrix contains 14,813 rows, 59 columns, 272 unique paper titles, and 310 unique source-file entries. A direct BSD-local search found no rows for the exact terms *Birch*, *Swinnerton*, *Mordell*, *Selmer*, *Shafarevich*, *Tate–Shafarevich*, *Mordell–Weil*, *elliptic curve*, *Tamagawa*, *Iwasawa*, *Heegner*, *Kolyvagin*, or *Gross–Zagier*. The string *BSD* occurred twice, but manual sampling did not identify a BSD-specific support arc. The scan found scattered L-function language and many generic rank/regulator rows, but no local Birch–Swinnerton-Dyer development.

This fact is not hidden. It fixes the proof posture. The present manuscript is a new BSD-local instantiation of the already-mature AASC role-compression machinery. The corpus matrix supplies generic structural support, not BSD-specific arithmetic claims.

Support family searched in matrix	Rows	Paper titles	Use here
Role-compression / role-separation / role-occupation terms	51	29	Template support for role architecture.
Bridge / bridge-completion / boundary trace terms	1,964	110	Template support for analytic-arithmetic bridge discipline.
UEAP / report discipline / carrier adequacy terms	894	45	Reportability and claim-standing support.
AMetric / selector / no-selector terms	3,025	171	No numerical or generator selector at rank boundary.

Closure / exhaustion / same-scope totalization terms	10,436	250	Route-locus exhaustion and no unbridged totalization.
No-generator / no-carrier / no-repair / silent-redescription terms	124	44	Blocking theorem family.

Remark 2.1 (Scan discipline)**.** The matrix scan is audit context, not a proof of BSD and not a mathematical premise about elliptic curves. The live proof chain below uses only local definitions and local restatements of AASC structural constraints.

2.2 Dependency ledger

This paper imports AASC results as fixed structural constraints. They are not imported to prove point existence, Sha finiteness, analytic continuation, or a leading-coefficient formula. They are imported to classify admissibility, standing, target fixation, reportability, bridge occupation, and illicit transition routes.

Imported result	Corpus statement	Local BSD form used here
Kernel theorem	Fixed-domain construction requires reference, standing, admissibility, and irreversibility.	A BSD rank claim must fix E, $L(E, s)$, n, the Mordell–Weil group, the rank-readout slot, and the bridge role before arithmetic rank can carry standing.
UEAP certification	Report status may not exceed target fixation, carrier adequacy, alternative exhaustion, laundering exclusion, and report preservation.	A report of $\operatorname{ord}_{s=1} L(E, s) = n$ may not be issued as $\operatorname{rank} E(\mathbb{Q}) = n$ unless the analytic-arithmetic bridge is discharged.
Structure of admissibility	Admissibility is AMetric, bivalent, unique on a fixed domain, and standing-conserving.	There is no graded, proximity, likelihood, numerical-confidence, or partial rank-readout status between analytic rank and Mordell–Weil rank on the same fixed report slot.
AMetric Boundary	No selector, ranking, metric, parameter, or discrete indexing can perform boundary work.	A numerical approximation, height search, preferred generator, Selmer threshold, or rank guess cannot select arithmetic rank before the bridge is fixed.
Same-scope operator closure	Partial same-scope operators cannot be totalized on the old carrier by domain enlargement, completion, repair, family change, or hidden auxiliary data.	The analytic-rank-to-arithmetic-rank interface cannot be totalized across all $E/\mathbb{Q}$ without bridge completion.

Impossibility Suite	No generators, no carriers, no repairs.	Mordell–Weil rank cannot be generated by analytic rank, carried from known cases or surrogate carriers, or repaired after unbridged reporting.
Standing-admissibility identity closure	No second same-domain standing classifier survives.	A same-scope non-Mordell–Weil rank classifier cannot govern the rank-readout slot if it disagrees with the free-rank status of $E(\mathbb{Q})$.
Reference fixation	Role assignment must be fixed before confirmation or readout.	The readout role $r_{\mathrm{rank}}(E, n)$ must be fixed before rank reportability; later point discovery or descent interpretation cannot fix the earlier report as the same act.
Boundary-trace fixation	A bridge trace must be declared, bivalent, stable, outcome-independent, and fail-closed.	The BSD bridge must be a declared analytic-arithmetic relation, not an oracle, approximation, scalar ranking, or late calibration.
Role-occupancy closure	Realization is compatible role occupation, not bare candidate existence.	Multiple bases of $E(\mathbb{Q})$ may occupy the same rank role; rank readout does not require a unique preferred basis.
No post-selection repair	Later filtering, selection, or correction cannot repair inadmissible generation as the same act.	Later discovery of rational points produces fresh bridge evidence; it does not retroactively make an earlier unbridged rank report admissible.
Prohibition of silent redescription	Untracked reference change cannot preserve standing.	Selmer groups, Sha data, Heegner points, regulators, numerical computations, or Iwasawa-theoretic structures cannot be silently redescribed as Mordell–Weil rank.
Non-factorizability of admissibility	Admissibility is global with respect to standing-bearing commitments and cannot be inferred merely by composing admissible parts.	Local factors, local heights, descent fragments, and local rank constraints do not automatically compose into global Mordell–Weil rank readout.

2.3 Local restatement: kernel and claim standing

Proposition 2.2 (Local kernel form). *A same-scope BSD rank claim has standing only when the following are fixed: the elliptic curve $E/\mathbb{Q}$, the integer n, the analytic L-function carrier, the Mordell–Weil group carrier, the requested report status, and the readout role being claimed. If any of these moves without explicit reset, the claim is not the same fixed-domain claim.*

Proof. Reference fixes the object of evaluation: (E, n) together with the relevant analytic and arithmetic carriers. Standing fixes the status being asserted: analytic vanishing order or arithmetic rank readout. Admissibility fixes which transformations preserve the domain and report slot.

Irreversibility blocks later repair of a failed report as the same report. Changing any member of the tuple changes the evaluated act or resets the domain. ☐

2.4 Local restatement: UEAP report discipline

Proposition 2.3 (Local UEAP form)**.** *A BSD report may be issued at arithmetic-rank status only if its carrier and bridge support arithmetic rank readout. An analytic carrier supporting* $\mathrm{Van}(E,n)$ *supports analytic vanishing-order status, not Mordell–Weil rank readout, unless the analytic-arithmetic bridge is occupied.*

Proof. UEAP separates target, carrier, alternatives, laundering, and reportability. The target of analytic vanishing-order standing is behavior of $L(E,s)$ at $s=1$. The target of arithmetic rank readout is the free rank of $E(\mathbb{Q})$. Since these targets differ, the stronger arithmetic report cannot be obtained by wording alone. It requires a bridge that aligns the analytic carrier with the Mordell–Weil rank slot. Without such alignment, the stronger report is a new claim. ☐

2.5 Local restatement: AMetric no-selector boundary

Proposition 2.4 (No selector at the rank-readout boundary)**.** *No selector, metric, ranking, canonical-generator rule, scalar strength, probability, typicality, numerical approximation, finite search height, or discrete index can decide Mordell–Weil rank readout on the pre-bridge side of the BSD claim.*

Proof. The arithmetic-rank boundary asks whether the readout role $r_{\mathrm{rank}}(E,n)$ is occupied by the free-rank structure of $E(\mathbb{Q})$. Any pre-bridge criterion that selects a rank value without being or supplying the Mordell–Weil rank bridge performs boundary work. The AMetric theorem excludes such selector or parameter work inside a fixed admissibility interface. If the criterion is declared as a new bridge, it must be typed and audited as such; otherwise it is inadmissible. ☐

2.6 Local restatement: same-scope operator closure

Proposition 2.5 (No unbridged totalization of the rank interface)**.** *The interface* $\mathrm{Van}(E,n)\mapsto \mathrm{RankRead}(E,n)$ *cannot be made universally valid over all elliptic curves by a same-scope move unless that move supplies bridge occupation. Domain enlargement, carrier collapse, evaluative completion, numerical extrapolation, descent substitution, and non-conservative family change are bookkeeping, explicit scope reset, or inadmissible unless they are bridge-complete.*

Proof. A totalization attempt must change at least one of the following: which curves are in domain, what counts as rank readout, what equivalence collapses readouts, what evaluator decides missing cases, or which family of objects is being used. Same-scope operator closure classifies these moves as bookkeeping, explicit scope reset, or inadmissible unless the missing datum is already standing-fixed. In BSD form the missing datum is the analytic-arithmetic bridge from vanishing order to Mordell–Weil rank. ☐

2.7 Local restatement: no generators, carriers, repairs

Proposition 2.6 (No generator, carrier, or repair of Mordell–Weil rank)**.** *No property* $Q(E,n)$ *distinct from the Mordell–Weil rank condition may generate, carry, or repair arithmetic rank readout. If* Q *authorizes rank readout, then either* Q *is extensionally bridge-complete for the rank condition, or it is a surrogate, scope reset, or inadmissible carrier attempt.*

Proof. If Q generates rank readout from analytic standing alone, it creates standing for the stronger report without the declared gate. If Q carries rank readout from another domain, it transfers standing across domains without fresh gating. If Q repairs an earlier unbridged report, it reverses failure for the same evaluated act. These are exactly the no-generator, no-carrier, and no-repair impossibilities. The only admissible case is that Q supplies the bridge itself. □

2.8 Local restatement: no silent redescription and non-factorizability

Proposition 2.7 (No silent surrogate occupation)**.** *A Selmer group, Tate–Shafarevich group, Heegner point package, regulator, height pairing, Iwasawa module, numerical L-function computation, or special-case theorem cannot occupy $r_{\mathrm{rank}}(E, n)$ unless it is bridged to the free-rank structure of $E(\mathbb{Q})$ for the same curve and the same rank value.*

Proof. The role $r_{\mathrm{rank}}(E, n)$ is typed by the condition $E(\mathbb{Q})/E(\mathbb{Q})_{\mathrm{tors}} \cong \mathbb{Z}^n$. A surrogate object lies in a different carrier unless a bridge maps it into that rank-readout condition. Treating the surrogate as if it already occupied the arithmetic-rank role changes the referent of the readout without explicit re-admission; this is silent redescription. If several local surrogate pieces are individually admissible, their assembly still requires a global bridge, because admissibility of the whole is not factorizable from admissibility of parts. □

2.9 Local restatement: role occupancy

Proposition 2.8 (Rank multiplicity is not basis multiplicity)**.** *If $E(\mathbb{Q})$ has rank $n > 0$, this does not create multiple rank-readout roles according to the many possible bases of its free part. It means the one role $r_{\mathrm{rank}}(E, n)$ is occupied. The standing-relevant question is whether the rank role is occupied, not whether a unique preferred basis of rational points has been selected.*

Proof. Role-occupancy closure separates representative multiplicity from target-role occupation. The target role is fixed by the claim: arithmetic rank readout for E with value n. Multiple Mordell–Weil bases may represent the same free abelian rank. Only a difference that changes the readout role, bridge condition, or report status creates a distinct standing-bearing occupation problem. □

2.10 Local replay of imported constraint results

Proposition 2.9 (Local no-generator replay)**.** *Analytic vanishing-order standing cannot generate Mordell–Weil rank readout.*

Proof. Analytic standing is the predicate $\mathrm{Van}(E, n)$. Mordell–Weil rank readout is the predicate $\mathrm{RankRead}(E, n)$. If the first generated the second without a bridge, then an analytic status would create arithmetic standing in a distinct carrier. That is standing generation. Under the no-generator principle, the only admissible way for analytic standing to yield arithmetic rank is for the bridge to be completed. □

Proposition 2.10 (Local no-carrier replay)**.** *Known cases, Selmer standing, Heegner standing, Iwasawa standing, regulator standing, numerical standing, or* Sha*-based standing cannot transfer into Mordell–Weil rank readout without target gating.*

Proof. Each listed structure lies in a carrier whose standing conditions differ from the free-rank structure of $E(\mathbb{Q})$. If its standing is used to authorize $\mathrm{RankRead}(E, n)$, then standing is being carried from one domain into another. The no-carrier principle permits no such transfer. The structure may become part of a bridge program, but it authorizes arithmetic rank readout only when it entails the actual rank condition. □

Proposition 2.11 (Local no-repair replay)**.** *Later discovery of rational points, later descent computation, or later proof of* Sha *finiteness creates a new bridge-complete report; it does not repair an earlier unbridged rank report as the same act.*

Proof. An unbridged report that E has rank n lacks the required bridge at the time of reporting. If later evidence supplies the missing bridge, the report now has additional certificate data. That is a fresh evaluated report with a different certificate. Treating the earlier unbridged report as having been standing-positive all along would retroactively change its status, which is post-selection repair. ☐

Proposition 2.12 (Local no-silent-redescription replay)**.** *A non-Mordell–Weil surrogate cannot be reported as arithmetic rank readout without changing the declared readout slot.*

Proof. The arithmetic-rank slot is typed by $E(\mathbb{Q})/E(\mathbb{Q})_{\text{tors}} \cong \mathbb{Z}^n$. A surrogate not satisfying that condition occupies a different role. Reporting it as rank readout changes the referent of the report while preserving the appearance of the same status. That is silent redescription unless an explicit bridge maps the surrogate into the rank condition. ☐

Proposition 2.13 (Local non-factorizability replay)**.** *Local BSD data do not compose into global Mordell–Weil rank readout without a global compatibility certificate.*

Proof. A collection of local Tamagawa factors, local heights, local solvability data, descent fragments, or finite checks does not by itself establish $E(\mathbb{Q})/E(\mathbb{Q})_{\text{tors}} \cong \mathbb{Z}^n$. The assembled object must still satisfy the global group-rank condition. If these interaction conditions are not checked globally, the report asserts whole-domain admissibility from admissible parts, which is precisely the factorization failure. ☐

2.11 Route-locus exhaustion for unbridged BSD readout

Scope of exhaustion. The theorem below is not a taxonomy of all arithmetic geometry and not a classification of every possible method for proving BSD. It exhausts unauthorized same-scope report transitions from analytic standing to Mordell–Weil rank readout inside the fixed AASC claim architecture. The exhaustion is over report-transition coordinates, not over mathematical proof strategies. A route that actually proves rank equality for the target curve is bridge-complete and is not blocked.

Definition 2.14 (Eight-coordinate normal form for unbridged BSD transition)**.** Let an attempted report begin with the analytic standing report

$$A_0(E, n): \quad \operatorname{ord}_{s=1} L(E, s) = n$$

and attempt to emit the arithmetic readout report

$$R_0(E, n): \quad \operatorname{rank} E(\mathbb{Q}) = n.$$

If no bridge-completion certificate $B^{\text{comp}}_{\text{BSD}}(E, n)$ is supplied, then the attempted transition can change only the following report coordinates: status, readout slot, rank-role occupation, carrier/domain, certificate time, local/global assembly, selection relation, or governing classifier. These are coordinates of the claim act, not examples of arithmetic methods. The normal form classifies where a report can change while claiming to preserve the same target, not how arithmetic geometers may attempt a proof.

Proposition 2.15 (Coordinate exhaustiveness)**.** *The eight coordinates of Definition 2.14 exhaust unbridged same-scope routes from analytic vanishing-order standing to Mordell–Weil rank readout.*

Proof. The target report $R_0(E, n)$ contains exactly a report status, a readout slot, an occupation condition for that slot, a carrier identity, a certificate time, a coverage domain, a selection relation if representatives are used, and a governing classifier. An unbridged route that produces a stronger report must alter at least one of these coordinates. If none is altered, the report remains analytic standing and has not become arithmetic rank readout. The proof therefore does not assert that BSD proof methods are exhausted; it asserts that unauthorized report transitions have only these coordinates available. Hence the coordinates are exhaustive for the fixed same-scope claim architecture. □

Theorem 2.16 (Route-Locus Exhaustion for Unbridged BSD Rank Readout)**.** *Every unbridged same-scope attempt to report Mordell–Weil rank readout from analytic vanishing-order standing must enter through one of eight loci: status promotion, readout-slot substitution, rank-occupation creation, carrier transfer, temporal repair, local factorization, selector authority, or second classifier. No ninth same-scope locus remains.*

Locus	BSD form	Blocking result
Status promotion	Analytic rank reported as Mordell–Weil rank.	UEAP / no unbridged promotion.
Readout-slot substitution	Selmer rank, Sha standing, regulator data, Heegner data, or numerical data replaces rank $E(\mathbb{Q})$.	Silent redescription.
Rank-occupation creation	The rank role is declared occupied without group-rank bridge.	No generators.
Carrier transfer	Known cases, Iwasawa results, modularity, or rank 0/1 results imported into a target curve without fresh gating.	No carriers.
Temporal repair	Later point construction or descent proof validates an earlier unbridged report as the same report.	No post-selection repair.
Local factorization	Local factors, local heights, or finite descent fragments treated as global rank readout.	Non-factorizability.
Selector authority	Numerical precision, search height, preferred generators, or height ranking used as gate.	AMetric boundary.
Second classifier	A rival same-domain rank classifier governs the Mordell–Weil rank slot.	Standing-admissibility identity closure.

Proof. Let an attempted report move from $\mathrm{Van}(E, n)$ to $\mathrm{RankRead}(E, n)$ without bridge occupation. If it changes report status while preserving the same carrier appearance, it is status promotion. If it replaces the Mordell–Weil rank slot with a Selmer, regulator, Sha, Heegner, numerical, or Iwasawa output, it is readout-slot substitution. If it declares rank occupation without proving the free-rank condition, it is rank-occupation creation. If it borrows standing from another theorem, family, or carrier, it is carrier transfer. If later data validate an earlier unbridged report as the same report, it is temporal repair. If local facts are assembled into global rank without a compatibility certificate, it is local factorization. If it chooses a rank value by numerical precision, search height, or preferred representation, it is selector authority. If a rival same-domain classifier governs the rank slot while claiming not to alter the target, it is a second classifier. By Proposition 2.15, these loci exhaust the ways an unbridged transition can alter status, readout slot, occupation, carrier/domain, certificate time, locality, selection, or classifier governance. Any route that alters none of these leaves the report unchanged and cannot produce arithmetic rank readout. A proposed ninth route must therefore either alter one of the eight coordinates under another name or else remain at analytic-standing status without producing Mordell–Weil rank readout. □

2.12 Dependency discipline

No imported result is used to show $\operatorname{rank} E(\mathbb{Q}) = n$ for a particular curve. The imported results are used to prove that only bridge-complete occupation of the Mordell–Weil rank role can license arithmetic rank readout, that no other same-scope route licenses that role, and that a classical proof would be exactly a proof of bridge completion. This is why the paper is a structural solution, not an arithmetic-geometric proof of BSD.

3 Classical BSD carrier, standing, and rank-readout interface

3.1 Elliptic curves and Mordell–Weil rank

Let $E/\mathbb{Q}$ be an elliptic curve. The Mordell–Weil group $E(\mathbb{Q})$ is a finitely generated abelian group. Its free rank is the integer r such that

$$E(\mathbb{Q}) \cong E(\mathbb{Q})_{\mathrm{tors}} \oplus \mathbb{Z}^r.$$

The arithmetic rank-readout predicate for E and n is

$$\mathrm{RankRead}(E, n) \iff r = n.$$

This is an arithmetic group-status predicate. It is not a statement about the order of vanishing of an analytic function.

3.2 The L-function and analytic rank

For an elliptic curve $E/\mathbb{Q}$, the analytic carrier used by BSD is the package

$$C_L(E) = \big(L(E, s), \Lambda(E, s), \text{Euler factors}, \text{functional equation}, s = 1, \sim_{\text{analytic}}\big).$$

For elliptic curves over $\mathbb{Q}$, analytic continuation and the functional equation are fixed in the standard arithmetic setting through modularity. This manuscript uses those facts only to fix the analytic carrier; no step below uses modularity to prove rank equality. The vanishing order at $s = 1$ is usually called the analytic rank:

$$r_{\mathrm{an}}(E) = \operatorname{ord}_{s=1} L(E, s).$$

In ordinary BSD terminology this number is analytic rank. In this AASC proof it is called *analytic vanishing-order standing* when the emphasis is on report status. For $n \in \mathbb{Z}_{\geq 0}$,

$$\mathrm{Van}(E, n) \iff r_{\mathrm{an}}(E) = n.$$

This is an analytic status. It does not by itself supply a basis of rational points, a free-rank certificate, or a descent computation.

3.3 Three-layer BSD bridge architecture

The BSD bridge is used locally in the three layers already introduced in Definition 1.13. The point of repeating them here is to keep the classical BSD carrier/readout interface explicit:

1. $B^{\mathrm{slot}}_{\mathrm{BSD}}(E,n)$ fixes the transition slot between $\mathrm{Van}(E,n)$ and $\mathrm{RankRead}(E,n)$.
2. $B^{\mathrm{adm}}_{\mathrm{BSD}}(E,n)$ fixes same-scope admissibility discipline for any route through that slot.
3. $B^{\mathrm{comp}}_{\mathrm{BSD}}(E,n)$ is the proof-grade completion of the bridge by analytic and arithmetic traces for the same target.

Only the third layer contains a proof-grade arithmetic free-rank trace. The first two layers type the transition and exclude unbridged substitutes; they do not assert rank equality.

Proposition 3.1 (Local bridge anti-circularity)**.** *At the local BSD carrier/readout interface, neither $B^{\mathrm{slot}}_{\mathrm{BSD}}(E,n)$ nor $B^{\mathrm{adm}}_{\mathrm{BSD}}(E,n)$ contains $\mathrm{RankRead}(E,n)$ by definition. The arithmetic conclusion appears only at $B^{\mathrm{comp}}_{\mathrm{BSD}}(E,n)$.*

Proof. This is the local form of Proposition 1.14. The slot fixes the typed position of the analytic-to-arithmetic transition. Admissibility fixes same-target route conditions. Completion supplies the proof-grade arithmetic trace. Collapsing any of these layers would either make BSD definitional or prevent incomplete bridge programs from being audited. □

3.4 Rank BSD, refined BSD, and structural BSD

The rank part, refined formula, and structural result are different layers.

BSD layer	Bridge target	Completion burden
Rank BSD	$\mathrm{ord}_{s=1} L(E,s) \leftrightarrow \mathrm{rank}\, E(\mathbb{Q})$	Prove equality of analytic rank and Mordell–Weil free rank for the fixed curve.
Refined BSD	Leading coefficient $\leftrightarrow$ arithmetic product	Prove rank equality, $\mathrm{Sha}(E)$ finiteness/order, regulator, period, Tamagawa, and torsion normalization.
AASC structural BSD	Carrier/standing/readout/bridge typing	Prove role compression, no unbridged promotion, surrogate exclusion, totalization block, and bridge-completion boundary.

3.5 The bridge and refined formula

The rank bridge is the typed interface between analytic rank and arithmetic rank:

$$B^{\mathrm{rank}}_{\mathrm{BSD}}(E,n) : \mathrm{Van}(E,n) \rightsquigarrow \mathrm{RankRead}(E,n).$$

The full formula bridge is stronger. It includes the rank equality together with the compatibility of the leading coefficient with the standard BSD factors:

$$B^{\mathrm{full}}_{\mathrm{BSD}}(E) : L^*(E,1) = \frac{\Omega_E \,\mathrm{Reg}_E \,|\,\mathrm{Sha}(E)|\prod_p c_p}{|E(\mathbb{Q})_{\mathrm{tors}}|^2}$$

when all terms are standing-fixed and the relevant finiteness and normalization conditions are lawfully discharged. Here $L^*(E,1)$ denotes the leading coefficient of $L(E,s)$ at $s=1$ in the standard BSD normalization.

Remark 3.2 (Rank bridge versus full formula bridge)**.** The rank bridge and full formula bridge are not identical. The rank bridge governs the transition from analytic vanishing order to Mordell–Weil rank. The full formula bridge adds leading-coefficient, regulator, period, Tamagawa, torsion, and Tate–Shafarevich data. A proof of the refined formula would include the rank bridge, but the structural rank analysis does not require pretending that every formula term is already rank readout.

Remark 3.3 (Normalization and finiteness discipline for the refined formula)**.** Formula (2) is used schematically in the standard BSD sense. The regulator has the usual rank-zero convention; the product of Tamagawa numbers is over the relevant bad primes; and the term $|\,\mathrm{Sha}(E)|$ presupposes the finiteness and order status needed for the refined formula to be a standing-bearing endpoint. These qualifications are not cosmetic: in AASC mode they are bridge-factor typing conditions.

3.6 Four BSD statuses

The following statuses must be kept distinct.

Status	Mathematical form	AASC role
Carrier fixed	C_E is declared	elliptic-curve carrier role
Analytic standing	$\mathrm{ord}_{s=1} L(E,s) = n$	vanishing-order standing role
Arithmetic readout	$\mathrm{rank}\, E(\mathbb{Q}) = n$	Mordell–Weil rank-readout role
Bridge architecture	$B^{\mathrm{slot}}_{\mathrm{BSD}}$, $B^{\mathrm{adm}}_{\mathrm{BSD}}$, and $B^{\mathrm{comp}}_{\mathrm{BSD}}$ are distinguished	analytic-arithmetic transition architecture
Full formula completed	BSD leading coefficient formula is established	refined bridge-completion role

The BSD rank conjecture is analytic-to-arithmetic totality of the third status over the second. The structural solution in this paper is the proof that the transition from the second to the third is role-compressed and bridge-governed.

3.7 Descriptive mismatch

The paper distinguishes arithmetic readout failure from illicit report promotion. If one establishes

$$\mathrm{ord}_{s=1} L(E,s) \neq \mathrm{rank}\, E(\mathbb{Q}),$$

then one has established a descriptive mismatch statement. AASC does not forbid such a statement. It assigns it an obstruction or mismatch report status. What AASC forbids is treating mismatch, surrogate standing, or neighboring data as a same-scope realization status competing with Mordell–Weil rank readout while preserving the same readout slot.

Proposition 3.4 (Mismatch is not selector data)**.** *The statement* $\mathrm{ord}_{s=1} L(E,s) \neq \mathrm{rank}\, E(\mathbb{Q})$*, considered as a mathematical mismatch statement, is not an imported selector and is not inadmissible. It becomes relevant to the* AASC *no-selector discipline only if it is promoted into an alternative same-scope readout role or used to replace the declared arithmetic-rank slot.*

Proof. Mismatch is a comparison between two fixed predicates. A fixed comparison may return either truth value without importing a selector. The selector problem arises only when one uses the negative result to alter the role architecture: for example, by replacing Mordell–Weil rank readout with analytic rank standing while claiming the same report status. That move introduces a second classifier or changes the declared consequence slot; the descriptive statement alone does neither. □

3.8 Classical endpoint as bridge completion

The rank part of BSD can now be expressed without role compression:

$$\forall E/\mathbb{Q}\,\forall n \in \mathbb{Z}_{\geq 0}, \quad \mathrm{Van}(E,n) \Rightarrow \mathrm{RankRead}(E,n),$$

with the implicit converse when rank equality is stated as equality of values. This is universal analytic-arithmetic bridge completion. A future classical proof would prove this assertion by arithmetic-geometric means. The present paper proves the structural role architecture in which such a proof would live.

4 Role separation

4.1 Carrier is not analytic standing

Lemma 4.1 (Carrier-standing distinction)**.** *The elliptic-curve carrier C_E is not the same role as analytic standing* $\mathrm{Van}(E,n)$*.*

Proof. The carrier C_E exists once the curve $E/\mathbb{Q}$ and its attached analytic and arithmetic structures are fixed. Analytic standing requires a particular integer n and the predicate $\mathrm{ord}_{s=1} L(E,s) = n$. One can fix the carrier without fixing n as the analytic rank. Therefore the carrier role and the analytic standing role are not identical. □

4.2 Analytic standing is not arithmetic readout

Lemma 4.2 (Standing-readout distinction)**.** *Analytic standing* $\mathrm{Van}(E,n)$ *is not the same role as arithmetic rank readout* $\mathrm{RankRead}(E,n)$*.*

Proof. Analytic standing is the statement that $L(E,s)$ has vanishing order n at $s = 1$. Arithmetic readout is the statement that the free abelian rank of $E(\mathbb{Q})$ is n. The first statement is formed in the analytic L-function carrier. The second is formed in the Mordell–Weil group carrier. Since the two statements have different carriers and different report conditions, they are distinct roles. □

4.3 Readout is not bridge

Lemma 4.3 (Readout-bridge distinction)**.** *The arithmetic readout status* $\mathrm{RankRead}(E,n)$ *is not identical to the bridge* $B_{\mathrm{BSD}}(E,n)$*.*

Proof. The readout status is the rank value of $E(\mathbb{Q})$. A bridge is the lawful transition condition connecting analytic standing to that readout. The bridge governs reportability; it is not itself the reported group-rank status. If bridge and readout were identical by definition, the structural theorem would be circular. □

4.4 Rank readout is not preferred-basis selection

Lemma 4.4 (Rank-basis distinction)**.** *The arithmetic readout status* $\mathrm{RankRead}(E, n)$ *is not identical to a chosen basis of* $E(\mathbb{Q})/E(\mathbb{Q})_{\mathrm{tors}}$*.*

Proof. The readout status is the free rank n. A basis is one representative way to exhibit a free rank when a free part is identified. There may be many bases related by $\mathrm{GL}_n(\mathbb{Z})$. The rank role is occupied by free-rank status, not by a unique preferred basis. Thus basis selection is not part of the BSD rank endpoint unless separately declared. □

Theorem 4.5 (BSD Role-Separation Theorem)**.** *For fixed* (E, n)*, the elliptic-curve carrier, analytic* L*-function carrier, vanishing-order standing, Mordell–Weil rank readout, analytic-arithmetic bridge, and preferred-basis data are pairwise distinct standing-bearing roles in the BSD* AASC *architecture in the sense that collapsing any adjacent pair changes reportability or admissibility conditions.*

Proof. The carrier-standing distinction, standing-readout distinction, readout-bridge distinction, and rank-basis distinction are proved in the preceding lemmas. Collapsing carrier into analytic standing would treat the existence of the curve and its L-function as if it already fixed a vanishing order. Collapsing analytic standing into rank readout would treat analytic rank as if it already supplied Mordell–Weil rank. Collapsing readout into bridge would make the proof circular. Collapsing rank into basis would import a selector not required by the classical conjecture. Each collapse changes what may be reported. Therefore the roles are distinct. □

Corollary 4.6 (No definitional BSD)**.** *The statement* $\mathrm{ord}_{s=1} L(E, s) = n$ *is not definitionally equivalent to* $\mathrm{rank}\, E(\mathbb{Q}) = n$*. If it were, the rank part of BSD would be a definition rather than a conjecture.*

Proof. By the Role-Separation Theorem, analytic standing and arithmetic readout are distinct roles. A definitional equivalence would collapse them and erase the bridge condition. That is exactly what the theorem forbids. □

5 No unbridged analytic-to-arithmetic promotion

The core transition in BSD is

$$\mathrm{Van}(E, n) \rightsquigarrow \mathrm{RankRead}(E, n).$$

The Role-Separation Theorem says this is not a same-role continuation. This section proves the first exclusion theorem: no same-scope report may promote analytic vanishing-order standing to Mordell–Weil rank readout without bridge occupation.

5.1 The promotion schema

Definition 5.1 (Unbridged analytic-to-arithmetic promotion)**.** An unbridged analytic-to-arithmetic promotion is a report transition of the form

$$\mathrm{ord}_{s=1} L(E, s) = n \rightsquigarrow \mathrm{rank}\, E(\mathbb{Q}) = n$$

issued on the same fixed BSD domain without supplying a bridge certificate, a proof of rank equality for the target curve, or an explicit scope reset.

Unbridged promotion is the status-promotion locus in the route-locus table. It is not the same as conjecturing that the two ranks are equal. A conjecture may be stated. A proof may be attempted. A bridge program may be declared. The illicit move is to report the stronger arithmetic readout status as already licensed by the analytic standing status.

5.2 Carrier adequacy and report preservation

Lemma 5.2 (Carrier inadequacy for unbridged rank readout)**.** *The analytic carrier supporting* $\mathrm{Van}(E, n)$ *is not adequate by itself to support* $\mathrm{RankRead}(E, n)$*.*

Proof. The carrier for analytic standing is $L(E, s)$ and its behavior at $s = 1$. Arithmetic rank readout requires the free-rank structure of $E(\mathbb{Q})$. The data required for the latter are not contained in the analytic predicate $\mathrm{ord}_{s=1} L(E, s) = n$ as a same-role fact. Thus the analytic carrier does not satisfy the carrier-adequacy test for arithmetic rank readout unless the analytic-arithmetic bridge is supplied. □

Lemma 5.3 (Report promotion)**.** *An unbridged promotion changes report status.*

Proof. The report $\mathrm{Van}(E, n)$ says that the L-function has analytic vanishing-order standing. The report $\mathrm{RankRead}(E, n)$ says that the Mordell–Weil group has arithmetic rank n. Since the Role-Separation Theorem separates these roles, the second report is a different status class. Issuing it on the basis of the first changes the report object rather than preserving it. □

5.3 No generator, no carrier, no repair

Lemma 5.4 (No generator route)**.** *Analytic vanishing-order standing cannot generate arithmetic rank readout as a same-domain operation.*

Proof. A generator route would take $\mathrm{Van}(E, n)$ and produce $\mathrm{RankRead}(E, n)$ as a consequence of analytic standing alone. Under the no-generators principle, standing cannot be created for a stronger arithmetic report by internal transformation of a distinct analytic status. The only admissible way for the transformation to succeed is to supply bridge completion. In that case the bridge is no longer absent. □

Lemma 5.5 (No carrier route)**.** *Arithmetic rank readout cannot be carried from a neighboring domain into* (E, n) *without fresh target gating.*

Proof. A carrier route would say that a neighboring structure, such as a known rank 0 case, known rank 1 case, Selmer computation, Iwasawa theorem, Heegner construction, or numerical family result, carries rank readout into the fixed target. Under the no-carriers principle, standing does not transfer across domains without fresh gating. The target (E, n) requires its own rank-readout bridge. If the neighboring structure supplies such a bridge, it functions as bridge completion. If not, it is carrier transfer and cannot authorize the readout. □

Lemma 5.6 (No repair route)**.** *An earlier unbridged rank report cannot be repaired later as the same report by later point discovery, descent computation, or formula proof.*

Proof. If the earlier report lacked bridge occupation, then it lacked arithmetic-rank standing at the time of assertion. A later discovery of rational points or a later proof may create a new bridge-complete report. It does not retroactively make the earlier unbridged report standing-positive as the same evaluated act. This is exactly no post-selection repair. □

Theorem 5.7 (No Unbridged Analytic-to-Arithmetic Promotion)**.** *No same-scope report may promote*

$$\operatorname{ord}_{s=1} L(E, s) = n$$

to

$$\operatorname{rank} E(\mathbb{Q}) = n$$

unless the arithmetic-rank role $r_{\text{rank}}(E, n)$ *is occupied by a lawful analytic-arithmetic bridge.*

Proof. By carrier inadequacy, the analytic carrier is inadequate by itself for arithmetic rank readout. By report promotion, the unbridged transition changes report status. By the no-generator, no-carrier, and no-repair lemmas, the transition cannot be supplied by generation, transfer, or repair. Therefore the only same-scope way to issue the arithmetic-rank report is to supply the bridge: a proof or certificate that lawfully connects analytic vanishing-order standing to Mordell–Weil rank for the same E and n. □

Corollary 5.8 (Conjecture and proof remain admissible)**.** *The theorem does not prohibit stating BSD or proving it analytically or arithmetically. A proof that establishes rank equality for all elliptic curves is a universal bridge-completion theorem, not an unbridged promotion.*

Proof. The forbidden move is reporting arithmetic rank without bridge occupation. A proof that establishes rank equality supplies the bridge. Hence it satisfies, rather than violates, the theorem. □

6 No surrogate arithmetic-rank readout

The second major escape route is surrogate substitution. Instead of proving Mordell–Weil rank, one may exhibit Selmer data, Heegner data, Tate–Shafarevich data, regulator data, numerical L-function data, or a special-case theorem, then report it as if it were rank readout. This section proves that such a move is inadmissible unless the surrogate is bridged to the free-rank structure of $E(\mathbb{Q})$.

6.1 Surrogate candidates

Definition 6.1 (Surrogate carrier)**.** A surrogate carrier for (E, n) is a structure $S(E, n)$ that supports a status related to rank but is not, as typed, the statement $E(\mathbb{Q})/E(\mathbb{Q})_{\text{tors}} \cong \mathbb{Z}^n$. Examples include Selmer groups, p-Selmer ranks, Tate–Shafarevich data, Heegner point packages, Euler-system data, regulator or height-pairing data, Iwasawa modules, numerical L-function computations, special-case rank theorems, and heuristic rank distributions.

A surrogate carrier may be mathematically valuable. It may be part of a bridge program. It may even be the correct route to a classical proof in a particular case. The point is narrower: it does not occupy $r_{\text{rank}}(E, n)$ unless it supplies or entails the Mordell–Weil rank condition.

6.2 The surrogate fork

Lemma 6.2 (Surrogate fork)**.** *Let* $S(E, n)$ *be a surrogate carrier. Exactly one of the following holds for arithmetic-rank purposes:*

1. $S(E, n)$ *supplies a bridge to* $\operatorname{RankRead}(E, n)$*, in which case it functions as bridge-completion support.*
2. $S(E, n)$ *does not supply such a bridge, in which case it does not occupy the arithmetic-rank role.*

Proof. The arithmetic-rank role is typed by $E(\mathbb{Q})/E(\mathbb{Q})_{\mathrm{tors}} \cong \mathbb{Z}^n$. If the surrogate supplies that condition, it has bridged into the readout role. If it does not, the role remains unoccupied. There is no third same-scope status between these alternatives because the admissibility interface is bivalent at the readout boundary: the rank role is occupied or not occupied relative to the certificate. ☐

6.3 Silent redescription

Proposition 6.3 (No silent surrogate readout)**.** *A surrogate carrier cannot be reported as Mordell–Weil rank readout unless an explicit bridge to* $\mathrm{RankRead}(E, n)$ *is supplied.*

Proof. Reporting a surrogate as rank readout changes the referent of the readout role. The declared readout role is free-rank status of $E(\mathbb{Q})$. A Selmer group, Sha object, regulator, Heegner package, Iwasawa module, or numerical computation is not automatically that free-rank status. Treating it as one without bridge changes the meaning of the report while claiming the same standing. That is silent redescription. The Prohibition of Silent Redescription blocks exactly this move. ☐

Theorem 6.4 (No Surrogate Arithmetic-Rank Readout)**.** *Selmer, Tate–Shafarevich, Heegner, Euler-system, regulator, height-pairing, Iwasawa-theoretic, numerical, modular, special-case, or heuristic structures do not occupy the arithmetic-rank role* $r_{\mathrm{rank}}(E, n)$ *unless they are bridged to the free-rank structure of* $E(\mathbb{Q})$ *for the same curve and the same rank value.*

Proof. Each listed structure is a surrogate carrier unless it produces or entails $\mathrm{RankRead}(E, n)$. By the Surrogate Fork, the unbridged case does not occupy the rank role. By No Silent Surrogate Readout, reporting the unbridged surrogate as rank readout is inadmissible. Therefore surrogate structures may serve as bridge programs or lower-status support, but not as arithmetic rank readout without bridge occupation. ☐

6.4 Consequences for standard language

The theorem requires careful wording. A phrase such as "Selmer rank suggests rank n" is not equivalent to "Mordell–Weil rank is n" unless the Selmer carrier is bridged to the Mordell–Weil rank condition. A phrase such as "Heegner construction supplies rank evidence" is not equivalent to "the target rank is globally read out" unless the construction proves the required rank statement. A phrase such as "numerical analytic rank n" is not equivalent to "arithmetic rank n" unless the numerical result is part of a certified bridge. The paper does not reject these routes. It prevents their overreporting.

7 No unbridged totalization of rank equality

The analytic-arithmetic BSD endpoint says that the analytic-rank interface is everywhere equal to Mordell–Weil rank over elliptic curves. In this paper the selector-free form is used: totality means bridge-complete rank equality for every target curve. This section proves that no unbridged same-scope totalization route is licensed.

Definition 7.1 (Rank-totality)**.** The BSD rank interface is total on elliptic curves over $\mathbb{Q}$ iff

$$\forall E/\mathbb{Q}, \quad \mathrm{ord}_{s=1} L(E, s) = \mathrm{rank}\, E(\mathbb{Q}).$$

This is exactly the classical rank endpoint. It is not rejected. The question is which routes can license such a status.

Definition 7.2 (Unbridged totalization attempt)**.** An unbridged totalization attempt is a same-scope move that treats the analytic-rank-to-arithmetic-rank interface as universally bridge-complete without producing bridge occupation for each E, without proving the universal rank equality, and without declaring a new target domain.

Lemma 7.3 (Totalization route exhaustion)**.** *Every unbridged totalization attempt falls into at least one of the following routes:*

1. *declaration of rank equality without proof;*
2. *selection of a rank value by numerical precision, search height, or heuristic distribution;*
3. *replacement of Mordell–Weil rank by Selmer rank or another surrogate;*
4. *repair of an unproved case by later data;*
5. *transfer from known cases or neighboring families;*
6. *composition of local factors, local heights, or descent fragments into global rank;*
7. *completion, modularity, Iwasawa, or model-theoretic existence used as a standing-extending operator;*
8. *explicit scope reset.*

Proof. To make rank equality universal in a report, a move must assert equality, choose a rank value, change what counts as rank, postpone or repair the rank status, import status from elsewhere, assemble it from parts, invoke a general operator family to create standing, or declare a new domain in which the target differs. These exhaust the locations at which totalization can enter the fixed claim: predicate, representative, codomain, time, domain, composition, operator family, or scope. □

Proposition 7.4 (Route disposition)**.** *The routes in the preceding lemma are either bridge completion, bookkeeping, explicit scope reset, or inadmissible. None supplies unbridged same-scope totalization.*

Proof. Declaration without proof is report promotion. Numerical rank selection is boundary selector import unless bridge-certified. Surrogate replacement is silent redescription unless bridged to Mordell–Weil rank. Later data produce fresh bridge evaluation, not same-act repair. Known-case transfer is carrier transfer unless the target curve is included in the proof. Local admissible pieces require a global compatibility certificate, because admissibility is non-factorizable. Completion, modularity, Iwasawa, and model-theoretic existence are same-scope operator families closed by exhaustion unless their output is already bridge-typed. Explicit scope reset is allowed, but it is not same-scope totalization. □

Theorem 7.5 (No Unbridged Same-Scope Totalization)**.** *The BSD rank interface cannot be made total over elliptic curves by unbridged same-scope totalization. Any admissible proof of rank equality is a bridge-completion theorem; any other route is bookkeeping, explicit scope reset, lower-status reportability, or inadmissible.*

Proof. By route exhaustion, any attempt to make the interface total enters through one of the listed routes. By route disposition, no route licenses unbridged same-scope totality. If a route supplies a proof of rank equality for each target curve, then it is no longer unbridged: it has completed the bridge. Therefore the theorem blocks only unbridged totalization, not classical proof. □

Corollary 7.6 (Analytic-arithmetic proof compatibility)**.** *A classical proof of the Birch and Swinnerton-Dyer Conjecture would count as universal bridge completion:*

$$\forall E/\mathbb{Q}, \quad \operatorname{ord}_{s=1} L(E,s) = \operatorname{rank} E(\mathbb{Q}).$$

It would not be an inadmissible totalization.

Proof. The theorem forbids unbridged totalization. A proof of rank equality supplies the bridge for each input. Thus it satisfies the required condition. □

8 Rank readout as role occupation

The role-occupancy layer prevents a second error: demanding a preferred basis of rational points. Mordell–Weil groups of positive rank do not come with a unique basis. The BSD target is not a canonical generator list. The target is occupation of the arithmetic-rank role.

Non-tautology notice. The role-occupation theorem below is not intended to compute rank. Its function is to type the readout role and to block three illicit replacements: analytic rank as arithmetic rank, Selmer or obstruction rank as arithmetic rank, and preferred-basis selection as rank. The theorem is a standing and reportability theorem, not a point-construction theorem.

Definition 8.1 (Arithmetic-rank role)**.** For fixed (E,n), let $r_{\mathrm{rank}}(E,n)$ denote the role "the arithmetic rank-readout slot for E has value n." The role is occupied iff $\operatorname{rank} E(\mathbb{Q}) = n$.

Definition 8.2 (Rank-role occupation set)**.** Let C be a fixed BSD certificate. The rank-role occupation set $\Sigma_C(E,n)$ consists of lawful assignments of the role $r_{\mathrm{rank}}(E,n)$ to the Mordell–Weil free-rank structure of $E(\mathbb{Q})$, subject to the certificate's declared equivalence, boundary, standing, and report rules.

Definition 8.3 (Lawful basis equivalence)**.** The group of lawful basis changes acts only by transformations preserving E, n, the Mordell–Weil group, torsion quotient, free rank, and report status. It may not erase rank-relevant distinctions or identify a surrogate with Mordell–Weil rank unless the certificate declares that bridge.

Lemma 8.4 (Wrong-rank case)**.** *If* $\operatorname{rank} E(\mathbb{Q}) \neq n$*, then* $\Sigma_C(E,n) = \varnothing$ *and arithmetic rank* n *is not occupied.*

Proof. An assignment occupying $r_{\mathrm{rank}}(E,n)$ requires the free-rank condition $E(\mathbb{Q})/E(\mathbb{Q})_{\mathrm{tors}} \cong \mathbb{Z}^n$. If the rank is not n, there is no lawful occupant for that role. Therefore the occupation set is empty. □

Lemma 8.5 (Correct-rank case)**.** *If* $\operatorname{rank} E(\mathbb{Q}) = n$*, then the arithmetic-rank role* $r_{\mathrm{rank}}(E,n)$ *is occupied. Multiple bases of the free part do not create multiple rank roles.*

Proof. If $\operatorname{rank} E(\mathbb{Q}) = n$, then $E(\mathbb{Q})/E(\mathbb{Q})_{\mathrm{tors}} \cong \mathbb{Z}^n$, which is exactly the rank-readout condition. If two bases are chosen, they are multiple representatives of the same free-rank status. The target role is still one: arithmetic rank readout of value n. □

Lemma 8.6 (Surrogate-exclusion case)**.** *A surrogate candidate not bridged to* $\operatorname{rank} E(\mathbb{Q}) = n$ *does not enter* $\Sigma_C(E,n)$ *as a live target-relative residue for arithmetic rank readout.*

Proof. The occupation set is typed by the arithmetic-rank role. A surrogate candidate lacking the Mordell–Weil free-rank condition fails eligibility for that role. If it is bridged into the rank condition, it is no longer merely surrogate. If it is not bridged, it is not live residue for $r_{\mathrm{rank}}(E,n)$. □

Theorem 8.7 (Mordell–Weil Rank as Role Occupation)**.** *For fixed (E, n) and certificate C, arithmetic rank readout is licensed precisely when the arithmetic-rank role $r_{\mathrm{rank}}(E, n)$ is lawfully occupied by the Mordell–Weil free-rank structure:*

$$\mathrm{RankRead}_C(E, n) \iff \operatorname{rank} E(\mathbb{Q}) = n.$$

After quotienting only lawful basis variation, the rank-readout role has one standing-bearing outcome when the rank is n and none when the rank is not n.

Proof. If $\operatorname{rank} E(\mathbb{Q}) \neq n$, the wrong-rank lemma gives no occupation. If $\operatorname{rank} E(\mathbb{Q}) = n$, the correct-rank lemma gives occupation. The surrogate-exclusion lemma ensures that no non-Mordell–Weil surrogate enters the role. Lawful quotienting removes only basis variation that preserves the rank status; it does not change whether the role is occupied. Therefore arithmetic rank readout is exactly Mordell–Weil rank-role occupation. □

Corollary 8.8 (No preferred-basis requirement)**.** *The structural solution does not require a canonical basis of $E(\mathbb{Q})/E(\mathbb{Q})_{\mathrm{tors}}$. It requires only lawful occupation of the rank role.*

Proof. The theorem defines readout by role occupation, not by unique basis selection. Multiple bases are compatible with the same readout role. □

9 The refined BSD formula as full bridge architecture

The rank part of BSD is only one layer. The refined formula adds a leading-coefficient bridge. This section records how the full formula is typed in the same architecture.

Definition 9.1 (Formula-factor roles)**.** For an elliptic curve $E/\mathbb{Q}$, the refined formula involves the following roles:

1. Ω_E: period/normalization role;
2. Reg_E: height-pairing and regulator role;
3. $\prod_p c_p$: local Tamagawa factor role;
4. $|E(\mathbb{Q})_{\mathrm{tors}}|$: torsion denominator role;
5. $|\,\mathrm{Sha}(E)|$: Tate–Shafarevich obstruction/completion role;
6. $L^{(r)}(E, 1)/r!$: analytic leading-coefficient role.

Proposition 9.2 (Formula factors are not rank readout)**.** *No individual formula factor occupies the Mordell–Weil rank-readout role. Each factor may be part of full bridge completion, but none is silently identical to* $\operatorname{rank} E(\mathbb{Q})$*.*

Proof. Each factor has its own carrier and report conditions. A period is an analytic normalization datum; a regulator is a height-pairing determinant; a Tamagawa number is local; a torsion factor is finite group data; $\mathrm{Sha}(E)$ is an obstruction/completion datum; the leading coefficient is analytic. The rank role is free-rank status of $E(\mathbb{Q})$. Treating any factor as rank readout without bridge changes the readout slot and is silent redescription. □

Theorem 9.3 (Full formula as refined bridge completion)**.** *The refined BSD leading-coefficient formula is a full bridge-completion architecture. It is admissible as a same-scope BSD endpoint only when the analytic leading-coefficient role, rank role, regulator role, period role, Tamagawa role, torsion role, and* Sha *role are all typed and lawfully connected.*

Proof. The formula equates an analytic leading-coefficient status with an arithmetic product of several distinct factors. Each factor has a distinct role. If the factors are typed and connected by proof, the formula is bridge-complete. If any factor is used to replace another without bridge, the result is silent redescription or carrier transfer. Therefore the refined formula is not a shortcut around role separation; it is the fully typed bridge architecture. □

10 Structural BSD Role-Compression Closure Theorem, AASC form

Non-tautology notice. The theorem below is not the tautology that rank may be reported once rank has been proved. Bridge completion is the classical arithmetic endpoint. The structural result is the theorem stack preceding it: BSD role compression, role separation, analytic-to-arithmetic no-promotion, surrogate-rank exclusion, no unbridged totalization, role-occupation discipline, and refined bridge typing. The arithmetic rank certificate is the bridge-completion condition, not the whole structural solution. Thus the theorem proves closure of the BSD role-compression architecture; it does not generate the arithmetic bridge.

The proof-order discipline of the paper is now complete. The BSD assertion has been defined as a role-compressed problem. The local AASC dependencies have been stated and locally replayed. The classical objects have been fixed. The roles have been separated. Unbridged promotion, surrogate readout, unbridged totalization, preferred-basis demand, and unlicensed route loci have been blocked. The structural solution theorem now follows.

10.1 Structural solution criterion revisited

Recall that $\mathrm{RoleSolution}_{\mathrm{AASC}}(P)$ requires five items: typed role separation, transition conditions, exhaustion of unlawful routes, bridge identification, and endpoint re-expression as bridge completion.

Lemma 10.1 (Typed role separation verified)**.** *The elliptic-curve carrier, analytic L-function carrier, vanishing-order standing, arithmetic rank readout, and analytic-arithmetic bridge are typed and separated for BSD.*

Proof. The BSD Role-Compression Lemma identifies the roles, and the BSD Role-Separation Theorem proves that they are distinct. The predicates Van and RankRead type the analytic standing and arithmetic readout. □

Lemma 10.2 (Transition condition verified)**.** *The transition from analytic standing to arithmetic rank readout is admissible only by bridge occupation of $r_{\mathrm{rank}}(E, n)$.*

Proof. The No Unbridged Analytic-to-Arithmetic Promotion Theorem proves that analytic standing alone cannot license arithmetic rank readout. The Rank-Occupation Theorem proves that arithmetic rank readout is role occupation. Therefore the transition condition is exactly bridge occupation. □

Lemma 10.3 (Unlawful-route exhaustion verified)**.** *The unlicensed routes from analytic standing to arithmetic rank readout are exhausted by the route loci of Theorem 2.16.*

Proof. The route-locus exhaustion theorem proves that any unbridged claim of arithmetic rank must alter status, readout slot, occupation, carrier, time, locality, selector authority, or classifier governance. If none is altered, the claim has not moved from analytic standing to arithmetic rank readout. If one is altered, the corresponding blocking theorem applies. Hence no ninth same-scope route remains. □

Lemma 10.4 (Bridge identification verified)**.** *The bridge required for the classical rank endpoint has three layers: the transition slot $B^{\mathrm{slot}}_{\mathrm{BSD}}(E,n)$, the same-scope admissibility discipline $B^{\mathrm{adm}}_{\mathrm{BSD}}(E,n)$, and the completion certificate $B^{\mathrm{comp}}_{\mathrm{BSD}}(E,n)$ that lawfully connects analytic vanishing-order standing to Mordell–Weil free-rank readout for the same E and n.*

Proof. By definition, $\mathrm{RankRead}(E,n)$ iff $\operatorname{rank} E(\mathbb{Q}) = n$. The rank-occupation theorem identifies this as role occupation. The no-promotion theorem identifies bridge occupation as the transition condition. The three-layer bridge definitions separate the typed slot, admissible same-scope routing, and classical completion certificate. Therefore the bridge is precisely lawful analytic-arithmetic connection between $\mathrm{Van}(E,n)$ and $\mathrm{RankRead}(E,n)$, with completion reserved for the classical arithmetic proof layer. □

Lemma 10.5 (Classical endpoint re-expressed)**.** *The unqualified BSD rank endpoint is universal bridge completion:*

$$\forall E/\mathbb{Q}\, \forall n \in \mathbb{Z}_{\geq 0}, \quad \mathrm{Van}(E,n) \Rightarrow \mathrm{RankRead}(E,n),$$

with equality of values expressed by the unique analytic rank value and the unique Mordell–Weil rank value.

Proof. This is the rank-totality form of the classical statement. □

Theorem 10.6 (Structural BSD Role-Compression Closure Theorem, AASC form)**.** *In* AASC *mode, the BSD role-compression problem is structurally closed as an analytic-arithmetic claim-architecture problem. More precisely,*

$$\mathrm{RoleSolution}_{\mathrm{AASC}}(\text{BSD-RoleCompression})$$

holds for the role-compression problem determined by the classical BSD assertion.

Proof. The BSD Role-Compression Lemma proves that the classical assertion compresses elliptic-curve carrier, analytic L-function carrier, vanishing-order standing, arithmetic rank readout, and analytic-arithmetic bridge. The Role-Separation Theorem proves that these roles are distinct. The no-unbridged-promotion theorem proves that analytic standing cannot be reported as Mordell–Weil rank without bridge occupation. The no-surrogate-readout theorem blocks Selmer, Sha, regulator, Heegner, Iwasawa, numerical, special-case, and heuristic carriers from silently occupying the rank role. The no-unbridged-totalization theorem exhausts selector, repair, transfer, completion, surrogate, numerical, descent, and local-factorization routes. The rank-occupation theorem identifies Mordell–Weil rank as role occupation rather than preferred-basis selection. The refined-formula theorem locates the leading-coefficient formula as full bridge architecture. These results satisfy the AASC structural solution criterion. Therefore the BSD role-compression problem is structurally solved in AASC mode. □

Corollary 10.7 (Classical bridge-completion corollary)**.** *A classical arithmetic-geometric proof of BSD would be a universal bridge-completion theorem inside the structure identified here. It would prove, for every $E/\mathbb{Q}$,*

$$\operatorname{ord}_{s=1} L(E,s) = \operatorname{rank} E(\mathbb{Q}),$$

and, for the refined conjecture, the leading-coefficient formula with all terms standing-fixed.

Proof. The structural solution identifies the bridge required for the transition from analytic vanishing-order standing to arithmetic rank readout. A classical proof supplies that bridge universally. Such a proof instantiates the structural architecture rather than competing with it. □

Remark 10.8 (Solution mode)**.** The theorem does not downgrade the classical BSD endpoint. It locates that endpoint. The structural solution closes the role-compression problem; the classical endpoint is the universal bridge-completion problem inside the closed structure. A reader demanding a classical arithmetic proof is asking for bridge completion, not for the structural result proved here.

11 Standard BSD routes as bridge tests

This section does not evaluate the analytic strength of the programs. It gives the AASC bridge test for each route. The discussion is not a dismissal of standard BSD machinery; it is a bridge audit. A route succeeds structurally only if it supplies the bridge from analytic standing to Mordell–Weil rank readout for the same elliptic curve.

11.1 Rank zero and rank one routes

Carrier supplied. Rank zero and rank one theorems provide bridge-complete information in their proved domains.

Readout pressure. They show how analytic information can become arithmetic rank readout when the target curve satisfies the hypotheses and the arithmetic conclusion is actually proved.

Bridge condition. The route must verify target inclusion, the analytic rank status, the arithmetic rank conclusion, and the obstruction controls required by the theorem invoked.

Failure if overreported. A domain-specific bridge-complete theorem is treated as universal BSD without target gating.

11.2 Gross–Zagier and Kolyvagin routes

Carrier supplied. Heegner point and Euler-system methods supply deep arithmetic data in specific hypotheses and are bridge-complete in the cases where their theorems prove the rank conclusion.

Readout pressure. They connect analytic derivatives and rational point information, especially in rank 0 and rank 1 contexts.

Bridge condition. The route must show that the target curve lies in the proved domain and that the construction supplies Mordell–Weil rank readout for that curve.

Failure if overreported. Local or special-case bridge completion is treated as universal BSD rank equality.

11.3 Selmer and descent routes

Carrier supplied. Selmer groups provide computable or partially computable structures that constrain Mordell–Weil rank and obstruction groups.

Readout pressure. Selmer rank bounds or approximates arithmetic rank once obstruction data are controlled.

Bridge condition. The route must bridge Selmer data to the actual free rank of $E(\mathbb{Q})$, including the relevant obstruction quotient.

Failure if overreported. Selmer rank is reported as Mordell–Weil rank without obstruction control.

11.4 Tate–Shafarevich group routes

Carrier supplied. Sha(E) carries obstruction/completion information in the refined BSD formula and in descent sequences.

Readout pressure. Its finiteness and order are central to the refined formula.

Bridge condition. Sha data must be lawfully integrated with Mordell–Weil rank and leading coefficient, not used as a rank substitute.

Failure if overreported. Obstruction-carrier standing is treated as rank readout.

11.5 Regulator and height-pairing routes

Carrier supplied. The regulator packages height-pairing data for independent rational points.

Readout pressure. A nonzero regulator certifies independence of a chosen rank-sized set under proper hypotheses.

Bridge condition. The height data must be attached to actual rational points and prove the required free rank; basis variation must be quotient-neutral.

Failure if overreported. Height data or a preferred height-minimal set is treated as canonical rank readout without a group-rank proof.

11.6 Iwasawa-theoretic routes

Carrier supplied. Iwasawa theory supplies p-adic analytic and arithmetic modules, main-conjecture bridges, and control theorems in suitable settings.

Readout pressure. These structures can connect p-adic L-functions, Selmer groups, and rank behavior.

Bridge condition. The route must bridge the p-adic carrier back to the classical BSD rank readout for the target curve.

Failure if overreported. p-adic standing is treated as classical Mordell–Weil rank without the control bridge.

11.7 Modularity and automorphic routes

Carrier supplied. Modularity supplies an automorphic carrier for $L(E, s)$ and its analytic structure.

Readout pressure. It authorizes analytic tools and functional equations.

Bridge condition. The automorphic carrier must connect analytic vanishing to rational-point rank; analytic continuation or functional equation alone is not rank readout.

Failure if overreported. Automorphic availability is treated as BSD completion.

11.8 Numerical verification and databases

Carrier supplied. Computations provide high-confidence evidence for individual curves or finite datasets, often including analytic rank guesses, descent data, and point searches.

Readout pressure. They are indispensable for exploration and can certify particular claims when the computation is proof-grade and all hypotheses are checked.

Bridge condition. Numerical data must be converted into certified analytic and arithmetic proof for the target curve; precision alone is not standing.

Failure if overreported. Approximate analytic rank or finite search height is reported as Mordell–Weil rank.

11.9 Heuristic and statistical routes

Carrier supplied. Heuristics and distribution models supply expectations about ranks, Sha, and family behavior.

Readout pressure. They organize conjectural landscape and predict density or distribution phenomena.

Bridge condition. The heuristic must be replaced by proof governing the actual target curve and readout role.

Failure if overreported. Statistical expectation is treated as rank readout.

11.10 Full leading-coefficient route

Carrier supplied. The refined formula supplies all bridge-factor roles at once.

Readout pressure. A proof of the full formula would give stronger data than the rank equality.

Bridge condition. Each factor must be standing-fixed; no factor may silently substitute for another; the rank value used in the derivative order must match Mordell–Weil rank.

Failure if overreported. Formula factors are used as independent generators of rank before the formula is bridge-complete.

11.11 Bridge-test summary table

Route	Carrier supplied	Bridge condition	Failure if overreported
Rank zero/one theorems	Proved special-domain arithmetic results	Target curve lies in the proved domain and the rank conclusion is established	Special case treated as universal
Gross–Zagier/Kolyvagin	Heegner-point and Euler-system data	Construction proves the Mordell–Weil rank for the fixed curve	Local bridge transferred globally
Selmer/descent	Descent and obstruction carrier	Selmer data bridge to actual free rank of $E(\mathbb{Q})$	Selmer rank reported as rank
Tate–Shafarevich data	Obstruction/completion carrier	Sha(E) is fixed and integrated into the target bridge	Obstruction reported as readout
Regulator/heights	Height-pairing carrier	Height data attach to actual rational points and certify free rank	Preferred basis treated as rank
Iwasawa/p-adic routes	p-adic analytic/arithmetic carrier	Control bridge returns to classical Mordell–Weil rank	p-adic standing as classical readout
Modularity/automorphy	Analytic L-function carrier	Analytic structure is connected to rational-point rank	Analytic carrier as rank proof
Numerical/data routes	Finite computational evidence	Proof-grade certificates replace approximation or search confidence	Evidence as theorem standing
Heuristics/statistics	Family-level expectation carrier	Heuristic is replaced by proof for the fixed curve	Expectation as readout
Refined formula	Full BSD factor architecture	Every factor is standing-fixed and the rank slot is occupied	Factors as rank generators

This table is a navigation device only. The detailed bridge tests above remain the operative statements; the table records the same route discipline in compressed referee-facing form.

12 What must be denied to reject the structural solution

A rejection of the structural solution must deny at least one of the following claims. This section records the exact pressure points.

Denial	AASC response
Analytic rank and Mordell–Weil rank are distinct roles.	If they were the same role, BSD would be definitional. The Role-Separation Theorem blocks this.
The classical BSD assertion compresses the analytic-to-arithmetic transition.	The BSD Role-Compression Lemma verifies the compression clause by clause.

Same-scope rank promotion requires a bridge.	UEAP report discipline, no generators, no carriers, and no repairs block unbridged promotion.
Surrogate carriers cannot silently occupy Mordell–Weil rank readout.	The no-surrogate theorem and prohibition of silent redescription block this.
Later arithmetic refinement cannot repair earlier unbridged reports as the same act.	The no-repair and no-post-selection-repair principles block retroactive standing.
Local analytic or arithmetic data do not factorize into global rank readout.	Non-factorizability requires a global compatibility certificate.
Rank role occupation is not preferred-basis selection.	The role-occupation theorem identifies rank readout as free-rank occupation, not basis preference.
The refined formula is bridge architecture, not a shortcut around role separation.	The full-formula theorem types every formula factor and forbids silent substitution.
Numerical precision or finite search can decide the rank slot.	The AMetric boundary blocks precision, search height, and preferred presentation as bridge authority unless converted into proof-grade certificate.
A second classifier may govern the same rank slot.	Standing-admissibility identity closure blocks rival same-domain rank governance.

Proposition 12.1 (No residual denial route)**.** *A denial of the structural solution must locate itself in one of the rows above or else reject* AASC *as a solution mode altogether.*

Proof. The theorem ladder covers role compression, role separation, rank-readout definition, no unbridged promotion, no surrogate readout, no unbridged totalization, role occupation, refined bridge architecture, and bridge completion. A denial of the solution must deny at least one of those steps. If it denies none, it concedes the structural solution. If it rejects the entire framework, the objection is framework-level rather than a defect in the BSD instantiation. □

13 Hostile-referee objections and replies

Audit 13.1 (This does not prove BSD arithmetically)**.** Correct. It is not an arithmetic-geometric proof of BSD. It is a structural AASC solution to the role-compression problem. An arithmetic proof would be bridge completion inside the structure.

Audit 13.2 (Analytic rank is legitimate data)**.** Correct. The paper preserves analytic vanishing-order standing. It blocks only unbridged promotion of analytic rank into Mordell–Weil rank readout.

Audit 13.3 (Selmer groups really do constrain rank)**.** Correct. They can be central bridge tools. The paper denies only that unbridged Selmer standing is itself Mordell–Weil rank readout.

Audit 13.4 (Gross–Zagier and Kolyvagin prove major cases)**.** Correct. In their proved domains, they are bridge-complete routes. The structural warning is against transferring those domains universally without target gating.

Audit 13.5 (The refined formula includes rank)**.** Correct. It includes rank as part of a larger bridge architecture. The formula does not license using any individual factor as a rank substitute before the full bridge is established.

Audit 13.6 (Numerical evidence can be proof-grade)**.** Sometimes, yes, when it is converted into certified analytic and arithmetic proof under the required hypotheses. The paper blocks precision or search height as an unlicensed selector, not certified computation as bridge support.

Audit 13.7 (The bridge hides the conclusion)**.** No. The bridge is layered. The slot and admissibility layers do not contain $\mathrm{RankRead}(E, n)$; they fix the typed transition and same-scope conditions. The completion layer contains the arithmetic readout because that is the classical BSD endpoint. If rank equality were built into the slot or admissibility layer, the theorem would be circular; this manuscript explicitly forbids that reading.

Audit 13.8 (The bridge requires the arithmetic conclusion, so the paper proves nothing)**.** No. The bridge-completion condition contains the arithmetic readout because that is exactly the classical endpoint. The paper does not claim to complete that bridge. It proves that, without such bridge completion, analytic rank, Selmer rank, numerical evidence, regulator data, Sha-status, p-adic standing, and local arithmetic data remain lower-status or surrogate reports. The result is a claim-status closure theorem, not an arithmetic generator.

Audit 13.9 (A future proof would refute this paper)**.** No. A future proof would instantiate the universal bridge-completion corollary. It would confirm the structural architecture by supplying the bridge universally.

Audit 13.10 (AASC solution mode is nonstandard)**.** This is a framework-level objection. The manuscript declares its mode and proves the theorem in that mode. A reader who rejects AASC as a solution mode rejects the framework, not the internal BSD instantiation.

Audit 13.11 (The theorem says only that rank readout requires rank readout)**.** No. The theorem stack proves role compression, role separation, no unbridged promotion, no surrogate substitution, route-locus exhaustion, no unbridged totalization, and role-occupation discipline. The result is not the tautology $R \Rightarrow R$, but the structural claim that the classical assertion compresses a bridge-governed transition from analytic standing to arithmetic readout.

Audit 13.12 (The BSD statement is already clear)**.** Correct as a classical arithmetic target. The claim here is different: the compact equality is not AASC structurally single-role, because it passes from analytic L-function standing to Mordell–Weil rank readout without displaying the bridge.

Audit 13.13 (Analytic rank already equals arithmetic rank under BSD)**.** BSD asserts that equality; it does not define analytic rank to be arithmetic rank. If the equality were definitional, BSD would not be an arithmetic conjecture.

Audit 13.14 (Iwasawa theory might prove BSD)**.** Yes. If an Iwasawa-theoretic route supplies the classical bridge for the target curve and readout role, it is bridge completion. The paper blocks only unbridged transfer from a p-adic carrier into the classical Mordell–Weil rank slot.

Audit 13.15 (The route list is only examples)**.** No. The list classifies structural coordinates of an unauthorized same-scope transition: status, readout slot, occupation, carrier, time, locality, selector, and classifier. It is not a catalogue of arithmetic methods.

Audit 13.16 (The matrix has no BSD-specific support)**.** Correct. This is declared as a source-freeze fact. The manuscript does not claim a prior BSD corpus theorem. It locally instantiates generic AASC machinery and carries the BSD-specific typing burden in-body.

Audit 13.17 (This belongs under arithmetic-geometry-only review)**.** Only if it is submitted as a classical BSD proof. As a structural AASC paper, the relevant review questions are: is the role compression real, are the roles separated, is the bridge non-circular, are unlicensed routes exhausted, and is the classical endpoint preserved as bridge completion?

14 Conclusion

The classical Birch and Swinnerton-Dyer Conjecture is classically well formed as an arithmetic-geometric mathematical proposition. In AASC mode, however, the compact assertion is structurally under-typed. It compresses elliptic-curve carrier, analytic L-function carrier, vanishing-order standing, Mordell–Weil rank readout, and analytic-arithmetic bridge into one apparent equality.

This paper has decompressed the assertion. It fixed the elliptic-curve and analytic carriers without building in rank equality, defined analytic vanishing-order standing separately from arithmetic rank readout, preserved descriptive mismatch reporting, and located the analytic-arithmetic bridge as the admissibility condition governing the transition from analytic rank to Mordell–Weil rank. It then proved that unbridged promotion, surrogate substitution, post-selection repair, silent redescription, carrier transfer, local-to-global factorization, and preferred-basis selection cannot supply same-scope rank readout.

The resulting theorem is a structural solution of the BSD role-compression problem in AASC mode. The BSD role-compression problem is structurally closed here as an analytic-arithmetic claim-architecture problem. This closure concerns the typed claim architecture and unauthorized report-transition coordinates; it is not a substitute for classical bridge completion. The classical arithmetic endpoint remains intact as the universal bridge-completion problem: a classical proof of BSD would show that every elliptic curve supplies the bridge required to occupy the Mordell–Weil rank-readout role with the analytic vanishing order and, in the refined version, would supply the leading-coefficient bridge factors lawfully. Such a proof would not compete with the structural solution; it would instantiate the bridge identified by it.

A Theorem ladder

Result	Function
BSD Role-Compression Lemma	Shows that the classical assertion compresses elliptic carrier, analytic standing, arithmetic readout, and bridge.
Laundering-risk theorem	Locates the risks created by leaving the roles compressed.
Matrix scan	Establishes proof posture: no BSD-local corpus arc; generic structural support only.
Dependency ledger	States imported AASC constraints in local BSD form.
Local replay lemmas	Make no-generator, no-carrier, no-repair, no-silent-redescription, and nonfactorizability claims explicit for BSD.
Route-locus exhaustion theorem	Proves that all unbridged routes enter through eight loci.
Role-separation theorem	Separates analytic carrier, vanishing standing, rank readout, bridge, and basis data.
No-unbridged-promotion theorem	Blocks reporting analytic rank as Mordell–Weil rank without bridge.
No-surrogate-readout theorem	Blocks Selmer, Sha, Heegner, regulator, Iwasawa, numerical, and heuristic substitutes absent bridge.
No-unbridged-totalization theorem	Blocks same-scope totalization by selector, repair, transfer, completion, surrogate, descent, or factorization.
Rank role-occupation theorem	Identifies rank readout as Mordell–Weil free-rank occupation, not basis preference.

Full formula theorem	Locates the refined leading-coefficient formula as full bridge architecture.
Structural BSD solution theorem	Concludes that the BSD role-compression problem is structurally solved in AASC mode as an analytic-arithmetic role-compression problem.
Classical bridge-completion corollary	Locates a classical proof as universal bridge completion.

B Claim-status ledger

Claim	Status in this paper	Classical proof burden preserved
$\mathrm{Van}(E, n)$	Analytic standing.	Establish analytic continuation and vanishing-order status as required in classical setting.
$\mathrm{RankRead}(E, n)$	Arithmetic readout.	Establish Mordell–Weil free rank.
$B_{\mathrm{BSD}}(E, n)$	Bridge condition.	Prove analytic-arithmetic transition for target curve.
Rank equality for all E	Universal bridge completion.	Full BSD rank proof.
Leading coefficient formula	Full bridge architecture.	Prove all formula terms and equalities standing-fixed.
Selmer data	Surrogate or bridge support.	Bridge to Mordell–Weil rank if used as readout.
$\mathrm{Sha}(E)$ data	Obstruction/completion role.	Prove finiteness/order and integration into formula if used.
Numerical data	Evidence or certified support.	Convert into proof-grade bridge if used as theorem standing.

C Formalization map

The following map gives a compact Lean-style shape for a future formalization of the structural layer. It is not a formal proof of classical BSD.

Manuscript object	Lean-style object
Elliptic curve carrier	`BSDCarrier`
Analytic vanishing standing	`VanishingStanding E n`
Mordell–Weil rank readout	`RankReadout E n`
BSD bridge slot	`BSDBridgeSlot E n`
BSD bridge admissibility	`BSDBridgeAdmissible E n`
BSD bridge completion	`BSDBridgeComplete E n`
Bridge anti-circularity	`bridgeSlotAndAdmissibilityDoNotContainRankReadout`
Role compression	`RoleCompresses BSD [carrier, Lcarrier, van, rank, bridge]`
No unbridged promotion	`noUnbridgedAnalyticToArithmeticPromotion`

No surrogate rank readout	`noSurrogateRankReadout`
Route-coordinate normal form	`RouteCoordinateNormalForm E n`
Route-locus exhaustion	`bsdRouteLocusExhaustion`
No ninth route	`noNinthUnbridgedRoute`
Rank occupation theorem	`rankReadoutRoleOccupation`
Full formula bridge	`bsdFullFormulaBridgeArchitecture`
Structural solution theorem	`structuralBSDSolutionAASC`
Bridge-completion corollary	`classicalProofIsBSDBridgeCompletion`

D Expanded audit capsules

D.1 Capsule 1: why analytic rank does not contain Mordell–Weil rank

Suppose analytic rank contained Mordell–Weil rank as part of its definition. Then the predicate $\mathrm{ord}_{s=1} L(E,s) = n$ would already entail $\operatorname{rank} E(\mathbb{Q}) = n$ by meaning. BSD would no longer be an arithmetic conjecture connecting analysis and rational points; it would be a definitional unpacking of "analytic rank." Since classical arithmetic geometry treats analytic rank and Mordell–Weil rank as distinct, AASC must type them separately.

D.2 Capsule 2: why the bridge cannot include the conclusion by definition

The bridge is the admissibility condition that licenses the transition. If the bridge were defined as "rank equality holds," then the structural theorem would hide the conclusion in the bridge. This manuscript forbids that reading. The bridge is a role: it is the typed place where a classical proof would supply the analytic-arithmetic connection.

D.3 Capsule 3: descriptive mismatch and counterexample status

A mismatch statement, if proven, is legitimate mathematical information. It would be a counterexample to the classical conjecture. AASC does not outlaw counterexamples by vocabulary. It blocks only the role-confused move of treating mismatch or analytic standing as if it were an alternative rank-readout role under the same target.

D.4 Capsule 4: Selmer rank as bridge support, not readout by default

Selmer groups are powerful because they sit near the Mordell–Weil group and obstruction data. That proximity does not erase role separation. A Selmer computation may be a bridge when it proves the target free rank. Without that bridge, Selmer standing remains a neighboring carrier.

D.5 Capsule 5: why numerical data require bridge certification

Numerical L-function evidence can be extremely valuable. The structural problem is not numerical work itself. The problem is using precision, finite search height, or database confidence as a boundary selector. Once numerical work is converted into a proof-grade certificate, it is bridge support. Before then, it is lower-status evidence.

D.6 Capsule 6: why rank readout is not basis selection

The free part of $E(\mathbb{Q})$ may admit many bases. The rank endpoint asks for the integer rank, not a canonical list of points. A proof may construct points to show rank at least n and use descent to show no more. But the readout role is rank occupation. Basis multiplicity is representative multiplicity, not role multiplicity.

D.7 Capsule 7: why Sha is bridge data, not rank readout

The Tate–Shafarevich group carries obstruction and completion information. It can be essential in a refined BSD bridge, but its role is not the free rank of $E(\mathbb{Q})$. Treating $\mathrm{Sha}(E)$ as rank readout silently replaces the target carrier.

D.8 Capsule 8: why local rank zero or one success does not transfer globally

A rank zero or rank one theorem supplies bridge completion only for curves satisfying its hypotheses. Transfer becomes illicit when a conclusion proved in one domain is reported for another curve without target inclusion and bridge verification.

D.9 Capsule 9: why Iwasawa standing requires classical bridge

Iwasawa theory may provide powerful p-adic standing. To occupy the classical BSD rank role, that standing must be bridged back to the Mordell–Weil rank of the fixed curve over $\mathbb{Q}$.

D.10 Capsule 10: why future BSD proof is bridge completion

A future arithmetic proof of BSD would discharge $B_{\mathrm{BSD}}(E, n)$ for every curve. It would instantiate the architecture isolated here rather than refuting it.

E Route-locus diagnostic worksheet

For any proposed BSD proof route, classify its decisive move:

1. Does it report analytic rank as Mordell–Weil rank? If yes, test status promotion.
2. Does it replace Mordell–Weil rank by Selmer, Sha, Heegner, regulator, Iwasawa, or numerical standing? If yes, test readout-slot substitution.
3. Does it declare the rank role occupied without a group-rank bridge? If yes, test rank-occupation creation.
4. Does it import standing from known cases, families, or neighboring theories? If yes, test carrier transfer.
5. Does it use later data to validate an earlier unbridged report? If yes, test temporal repair.
6. Does it assemble local data into global rank without compatibility? If yes, test local factorization.
7. Does it choose by precision, height, preference, or finite search? If yes, test selector authority.
8. Does it install a rival rank classifier while preserving the same report slot? If yes, test second classifier.

If none of these occurs and the route proves rank equality for the target curve, it is bridge-complete.

F Corpus support scan appendix

The project matrix scan produced the following direct-target result:

Search term	Matrix hits
Birch	0
Swinnerton	0
BSD	2
Mordell	0
Mordell–Weil	0
Selmer	0
Shafarevich	0
Tate–Shafarevich	0
elliptic curve	0
L-function / L function	29

The absence of BSD-local rows is a source-governance constraint. It prevents this manuscript from claiming to synthesize an existing BSD corpus arc. The manuscript instead states a new local BSD instantiation of generic AASC machinery. That is why all BSD-specific definitions are given locally before the main theorem.

G Wrong-mode boundary and final admissibility audit

G.1 Wrong-mode objections

The following objections are mathematically legitimate in the classical BSD proof class but do not refute the theorem proved here:

1. Demand a construction of rational points for every analytic-rank instance.
2. Demand a proof of finiteness of Sha(E).
3. Demand the full leading-coefficient formula.
4. Demand a new Euler-system, descent, modular, or Iwasawa theorem.
5. Demand that AASC generate missing arithmetic estimates.

Each demand asks for bridge completion. The present theorem identifies the bridge architecture and excludes substitutes; it does not supply the classical bridge.

G.2 Right-mode objections

A same-mode objection must attack at least one of the following:

1. the BSD assertion really does not move between analytic and arithmetic roles;
2. analytic rank and Mordell–Weil rank are not distinct report statuses;
3. the bridge slot, bridge admissibility, and bridge completion layers fail to separate circularity from endpoint completion;

4. an unbridged promotion route survives outside the route-coordinate normal form;
5. a listed surrogate really occupies Mordell–Weil rank without bridge;
6. rank readout requires a preferred basis rather than a free-rank role;
7. route-locus exhaustion misses a report coordinate;
8. a future classical proof would not be bridge completion inside the separated architecture.

These are the review targets for the structural theorem.

G.3 Final admissibility audit

The final audit has twelve checks:

1. **Classical endpoint check:** the BSD rank endpoint is stated as classically well formed.
2. **Claim-class check:** BSD role compression is distinguished from classical BSD truth.
3. **Compression check:** role compression is defined neutrally before use.
4. **Role check:** analytic rank is not built into Mordell–Weil rank.
5. **Bridge-layer check:** bridge slot, bridge admissibility, and bridge completion are separated.
6. **Mismatch check:** descriptive mismatch is preserved as legitimate mathematical data.
7. **Surrogate check:** Selmer, Sha, regulator, Iwasawa, numerical, and Heegner data are retyped, not dismissed.
8. **Known-case check:** rank 0/1 and other special results are treated as local bridge completions, not universal transfers.
9. **Route-locus check:** route exhaustion is limited to unbridged report transitions, not all arithmetic geometry or all proof strategies.
10. **Future-proof check:** a future arithmetic proof instantiates, not refutes, the bridge architecture.
11. **Source-use check:** BSD sources and AASC sources are separated by proof role.
12. **Endpoint notation check:** the final theorem states $\mathrm{RoleSolution}_{\mathrm{AASC}}(\text{BSD-RoleCompression})$, not a classical proof of BSD.

H Source and dependency ledger

Source class	Role in this manuscript	Load-bearing status
Hodge structural solution template	Provides model of carrier/standing/readout/bridge separation and role occupation without unique representative selection.	Template only; locally retyped.

RH structural solution template	Provides model of analytic carrier, descriptive-data preservation, anti-circular bridge definition, and route audit.	Template only; locally retyped.
AASC corpus matrix	Provides support scan and generic structural family counts.	Audit context.
AASC kernel / UEAP / AMetric / closure / impossibility results	Provide structural constraints restated locally.	Structural premises.
Standard elliptic-curve arithmetic	Supplies notation for $E(\mathbb{Q})$, rank, $L(E,s)$, Selmer, Sha, regulator, Tamagawa factors, and BSD formula.	Background notation only.
Classical BSD proof programs	Used as bridge-test examples, not as proof premises.	Non-load-bearing examples.

I Source-use audit

I.1 Why BSD-local matrix support is not required for this proof class

The corpus matrix contains no prior BSD-local theorem arc. That is not a hidden premise failure because no prior BSD theorem-content is used. Classical BSD sources fix the arithmetic objects and theorem vocabulary. The BSD-local burden–carrier, standing, readout, bridge slot, bridge admissibility, bridge completion, route-locus normal form, and role occupation–is carried by definitions and lemmas inside this manuscript. The imported AASC corpus machinery supplies only generic fixed-domain report constraints: no unbridged promotion, no carrier transfer, no silent redescription, no post-selection repair, no selector boundary crossing, and no nonfactorizable local-to-global report. The absence of BSD-local matrix rows is therefore a source-governance limitation, not a proof gap for the declared structural proof class.

Source class	Supplies	Does not supply
Classical BSD sources	Elliptic-curve vocabulary, analytic rank, arithmetic rank, refined-formula notation, and standard theorem landscape.	AASC role-compression criterion or the structural solution.
Gross–Zagier / Kolyvagin sources	Examples of local bridge completion in rank 0/1 settings and Euler-system contexts.	Universal BSD or automatic transfer to arbitrary curves.
Selmer and Sha sources	Descent and obstruction vocabulary.	Unbridged Mordell–Weil rank readout.
Iwasawa-theoretic sources	p-adic analytic/arithmetic bridge programs.	Classical rank readout unless the control bridge is discharged.
Numerical BSD sources	Evidence, examples, and proof-grade computation when certified.	Universal bridge completion by precision alone.
Clay BSD statement and surveys	Official problem framing and standard overview.	Any AASC proof step or bridge completion.

AASC corpus sources	Report-status, admissibility, no-selector, no-repair, no-surrogate, and role-occupation constraints.	Arithmetic proof of BSD.
Corpus matrix scan	Source-freeze and generic support profile.	Prior BSD-specific theorem arc.

J Additional bridge-test certificates

J.1 Certificate 1: Selmer route

A Selmer route is bridge-complete for rank only when the exact sequence and obstruction data force the Mordell–Weil free rank. Otherwise it is a neighboring carrier.

J.2 Certificate 2: Heegner / Euler-system route

A Heegner or Euler-system route is bridge-complete in the target domain when it proves the needed point existence, independence, and upper-bound data for the same curve. Outside that domain it is carrier transfer.

J.3 Certificate 3: numerical route

A numerical route is bridge-complete only after the computation is converted into proof-grade analytic and arithmetic certificates. Precision and finite search height are not bridge authority by themselves.

J.4 Certificate 4: refined formula route

A full formula route is bridge-complete only when all formula-factor roles are standing-fixed and the derivative order used in the leading coefficient is the Mordell–Weil rank established by the bridge.

References

[1] B. J. Birch and H. P. F. Swinnerton-Dyer, Notes on elliptic curves. I, *J. Reine Angew. Math.* 212 (1963), 7–25.

[2] B. J. Birch and H. P. F. Swinnerton-Dyer, Notes on elliptic curves. II, *J. Reine Angew. Math.* 218 (1965), 79–108.

[3] J. W. S. Cassels, Arithmetic on curves of genus 1. VIII. On conjectures of Birch and Swinnerton-Dyer, *J. Reine Angew. Math.* 217 (1965), 180–199.

[4] J. Tate, On the conjectures of Birch and Swinnerton-Dyer and a geometric analog, *Seminaire Bourbaki* 306 (1965/66).

[5] J. Tate, The arithmetic of elliptic curves, *Invent. Math.* 23 (1974), 179–206.

[6] J. H. Silverman, *The Arithmetic of Elliptic Curves*, Graduate Texts in Mathematics 106, Springer, second edition, 2009.

[7] J. H. Silverman, *Advanced Topics in the Arithmetic of Elliptic Curves*, Graduate Texts in Mathematics 151, Springer, 1994.

[8] J. E. Cremona, *Algorithms for Modular Elliptic Curves*, Cambridge University Press, second edition, 1997.

[9] B. Gross and D. Zagier, Heegner points and derivatives of L-series, *Invent. Math.* 84 (1986), 225–320.

[10] V. A. Kolyvagin, Finiteness of $E(\mathbb{Q})$ and $\mathrm{Sha}(E, \mathbb{Q})$ for a subclass of Weil curves, *Math. USSR Izvestiya* 32 (1989), 523–541.

[11] J. Coates and A. Wiles, On the conjecture of Birch and Swinnerton-Dyer, *Invent. Math.* 39 (1977), 223–251.

[12] A. Wiles, Modular elliptic curves and Fermat's last theorem, *Ann. of Math.* 141 (1995), 443–551.

[13] R. Taylor and A. Wiles, Ring-theoretic properties of certain Hecke algebras, *Ann. of Math.* 141 (1995), 553–572.

[14] C. Breuil, B. Conrad, F. Diamond, and R. Taylor, On the modularity of elliptic curves over $\mathbb{Q}$: wild 3-adic exercises, *J. Amer. Math. Soc.* 14 (2001), 843–939.

[15] J. S. Milne, *Arithmetic Duality Theorems*, Perspectives in Mathematics 1, Academic Press, 1986.

[16] C. Skinner and E. Urban, The Iwasawa main conjectures for GL_2, *Invent. Math.* 195 (2014), 1–277.

[17] M. Bhargava and A. Shankar, Binary quartic forms having bounded invariants, and the boundedness of the average rank of elliptic curves, *Ann. of Math.* 181 (2015), 191–242.

[18] Clay Mathematics Institute, The Birch and Swinnerton-Dyer Conjecture, Millennium Prize Problem official problem description.

[19] W. Stein and C. Wuthrich, Algorithms for the arithmetic of elliptic curves using Iwasawa theory, in *Algorithmic Number Theory*, Lecture Notes in Computer Science 5011, Springer, 2008.

[20] A. J. Maley, *A Structural Solution to the Hodge Conjecture as a Role-Compression Problem*, June 2026.

[21] A. J. Maley, *A Structural Solution to the Riemann Hypothesis as a Carrier-Preservation Role-Compression Problem*, June 2026.